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Evolvability and Organismal Architecture

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THE BLIND WATCHMAKER AND THE REMINISCENT ARCHITECT

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the Blind Watchmaker and the Reminiscing Architect

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We (the indivisible divinity that works in us) have dreamed the world. We have dreamed it resistant, mysterious, visible, ubiquitous in space and firm in time, but we have allowed slight, and eternal, bits of the irrational to form part of its architecture so as to know that it is false.

Jorge Luis Borges, Avatars of the Tortoise

Blindness has not been for me a total misfortune; it should not be seen in a pathetic way. It should be seen as a way of life: one of the styles of living.

Jorge Luis Borges, Blindness

1

INTRODUCTION

Stefano TISO

...if one could understand a flower as it has its Being in God
this would be a higher thing than the whole world!

attributed to *Meister Eckhart*

↓

Little flower—but if I could understand
What you are, root and all, and all in all,
I should know what God and man is

Alfred Tennyson, Flower in the Crannied Wall

↓

To see a World in a Grain of Sand.
And a Heaven in a Wild Flower.
Hold Infinity in the palm of your hand.
And Eternity in an hour.

William Blake, Auguries of Innocence

↓

...if we could understand a single flower we would know who we are and what the world is

Jorge Luis Borges, A Personal Anthology

1.1. TENETS

More than three and half billion years ago, while Earth's molten crust was still cooling down, life took root on this planet, and to this day, it is still flourishing. In this dynamic world where astronomical, geological, and ecological phenomena lead to ever-shifting conditions, living beings face perpetually changing trials. Their only way to survive and thrive is to adapt through the process of evolution by natural selection. Life and Evolution are thus the two sides of the same coin since their inception. For this reason, I argue that living beings throughout their long history must have, to an extent, evolved the ability to evolve.

Like other scientists, I, therefore, consider "evolvability", this capability to undergo adaptive evolution, a fundamental driver of the success or downfall of biological lineages. Yet, standard evolutionary theory has for a long time overlooked this topic, focusing on a more short-term view of evolution. The long-term influence of evolvability has been dismissed as negligible since, as Richard Dawkins eloquently expressed, evolution is considered akin to "a blind watchmaker".

In this thesis, I question some of the assumptions of standard evolutionary biology. I try to show that, when these assumptions are relaxed, evolution does not seem so blind, and that evolvability plays an important role, revealing a "reminiscing architect" next to the "blind watchmaker". Building upon past successes, evolvability empowers organisms to meet ever-new challenges. The solutions to past adversities remain ingrained in the architecture of those who survive, making them more prepared for future struggles. With only a few additions to the mainstream perspective, we can arrive at new conclusions; integrating evolvability and its consequences in a comprehensive evolutionary framework. These additions are:

- To consider evolution as taking place in a dynamic environment, where selective targets are never static but constantly changing. Organisms, therefore, do not adapt to a single and unchanging selective target but evolve adaptive response strategies to a series of challenges.

- To see organisms not as a sum of independent traits determined by independent genes, but as the emergent result of the coordinated action of many interconnected components. These components do not individually contribute to the organism's fitness, but structured together they build adaptive response mechanisms.

- Finally, and most importantly, to recognise that a fundamental determinant of evolvability is the structural relationship between the mechanisms that give rise to an organism i.e.: organismal architecture. It is in the architecture of organisms that we can read the records of the steps they took to increase their evolvability.

In the following introduction, I will expand on and articulate this brief declaration of intent. Firstly, I will elucidate the significance of the concept of evolvability and provide an overview of its many definitions and key features. Secondly, I will critically evaluate the limitations associated with the standard theoretical approach to studying evolvability. In doing so, I will highlight the need for a more nuanced and comprehensive understanding of this concept. Thirdly, I will present my rationale for exploring the role of architecture in facilitating a deeper understanding of evolvability. Specifically, I will argue that studying architecture can offer unique insights into higher-level evolutionary dynamics and can help shed light on the fundamental principles that govern the evolution of complex biological systems. Fourthly, I will explain the methodology that I intend to use to show the relevance of architecture and evolvability, providing a detailed justification for its suitability. Finally, I will give a brief outline of the research chapters contained in this thesis and their significance to my overall goal.

1.2. EVOLVABILITY

One of the most striking aspects of life on Earth is the remarkable diversity and success of different organisms, ranging from microscopic bacteria to complex humans. Despite their vast differences, many types of organisms have managed to thrive and avoid extinction. What sets successful organisms apart from those that go extinct? For instance, how can we explain the almost complete absence of multicellular organisms that exclusively use asexual reproduction (only one animal clade, Bdelloid rotifers, seems to exclusively display this reproductive mode) [1–3] ? This question puzzles biologists, as asexual reproduction has clear short-term advantages in replication efficiency. This question is one of the greatest dilemmas in evolution. One explanation, for instance, is that the persistence of multicellular sexual reproduction might be a necessary requirement for multicellular organisms to escape the constant extinction threat of unicellular parasites [4]. Although many multicellular organisms almost exclusively reproduce asexually they can switch to sexual reproduction under stressful conditions. This way they can respond by recombining their genetic material and find new solutions that allow them to survive. The scarcity of obligate asexual reproduction in multicellular organisms can thus, at least in part, be explained by the fact that many lineages that were obligatory asexual might have been wiped out from the evolutionary record due to a fatal parasitic invasion.

Sexual reproduction could not only help escape the parasitic onslaught but could also grant other selective advantages. For instance, it is also proposed it could help avoid other deleterious factors such as mutational meltdown, which happens in asexual populations and would not allow sustained survival of organisms of a certain complexity [5]. Thus, while less efficient at replication than asexuality in the short-term, given the right initial conditions (heterogeneous environment or selective pressures, small population and/or frequent bottlenecks) sexual reproduction can persist long enough to then triumph over asexuality thanks to its long-term benefits [1, 6].

On the other hand, one might ask: How can bacteria be able to survive their own parasites (i.e. bacteriophages) if they are asexual? And indeed it might very well be that they just developed a different design, which allows them to employ a strategy that differs from that of multicellular organisms. Their simpler and streamlined design makes them more robust to abrupt genetic changes and thus they are able to use other, somehow more radical, mechanisms for genetic recombination such as: integrating random DNA fragments picked up from the environment (transformation [7]), transferring plasmids to one another, even across very different bacterial families (horizontal gene transfer or conjugation [8]), and even by using some of their virus parasites themselves as vectors for exchanging genetic materials (transduction [9]). This is a clear example of how certain designs can prove more advantageous than others in favouring adaptive evolution under certain conditions (e.g. sexual reproduction for complex multicellular organisms; transduction, conjugation, and transformation for the more streamlined bacteria), and that organismal designs are driven by the necessity of being able to adapt to constantly new selective challenges.

An organism's ability to adaptively respond to new conditions is crucial for its survival and propagation. However, it is not trivial to explain how the capability to produce these adaptive responses evolved in the first place. Selection is blind, acting only on present, short-term conditions and variation, and cannot target features that facilitate long-term evolution to conditions that have not yet been met. Nevertheless, l will argue in this thesis that long-term features can and did evolve along evolutionary history. Lineages that accumulate short-term adaptations that eventually lead to long-term advantages will be able to produce better-matched phenotypes to new conditions. Thus, these serendipitously long-term oriented lineages will be more likely to persist and dominate the evolutionary record compared to those lineages whose short-term adaptations did not confer them the ability to respond adaptively to unforeseen conditions. For instance, when talking about sexual reproduction it has been shown that it is advantageous to use it only in a restricted set of circumstances [10, 11], but those few multicellular organisms who luckily acquired it because of the initial short-term need for it, and then managed to maintain it, were also those that to this day have been able to survive catastrophic events such as new pathogens outbreaks and long term genetic melt-down. The ability to maintain and gather long-term adaptation acquired through short-term selection is, in my opinion, greatly influenced by how organisms structure the relationship among their different traits, i.e. their architecture.

For this reason, I consider evolvability and evolvability-enhancing architecture fundamental to understanding long-term evolutionary dynamics, the factors that govern the success or failure of biological designs, and the evolution of adaptive responses. By including evolvability-enhancing architecture as a relevant factor, the framework of evolutionary theory is expanded. While standard evolutionary theory mainly focuses on single adaptation events through gradual changes it often treats long-term evolutionary phenomena as simple statistical events dictated by speciation and extinction rates. Standard theory to this day struggles to provide a proximate explanation of how microevolution leads to macroevolution [12, 13]. Evolvability allows for a description of the different strategies and constraints that different organisms have evolved while faring through ever-changing conditions and therefore helps explain the emergence and disappearance of groups of organisms (and their design) over time.

Evolvability can be defined broadly as ä system's capability to undergo adaptive evolution"(where capability here indicates the degree to which an organism is capable of undergoing adaptive evolution, not the presence/absence of this ability). However, as Pigliucci [14] argues in his opinion paper "Is evolvability evolvable?", evolvability encompasses a family of related but distinct phenomena. When discussing evolvability, we are referring not to a single concept but rather to overlapping ideas that illuminate different facets of evolution in action. Therefore a general definition has limited utility for focused research questions, as this concept has a very broad range of potential applications and spans multiple fields of research. As a result, scientists have proposed more specific definitions tailored to their areas of study.

Pigliucci identifies three different types of definitions based on the different evolutionary levels/scales on which they focus: evolvability as the genetic variation within a population which can act as the raw material for selection (genetic scale) [15], evolvability as the ability of an organism to produce variation, i.e. variability (genotype-to-phenotype scale) [16], and finally, evolvability as the capacity to undergo major evolutionary transitions (macroevolutionary scale) [17]. Other definitions have been proposed in evolutionary biology (see [18, 19], and Box 1 of Chapter 1 of this thesis), but they more or less adhere to the same pool of ideas. On the other hand, different fields of study have come up with definitions on their own. These fields "cross-fertilize" with biology, while, on the other hand, pursuing their own research interest, providing different definitions of evolvability. For instance, Charles Ofria, Christoph Adami, and other students of Artificial Life systems such as Avida, often define evolvability in terms of the ability to perform different or more complex (algorithmic) operations [20–22]. The goal of this research is to see if organisms can be seen in terms of algorithmic logic, and therefore if this type of logic could help us through its principle to better describe and reproduce biological life and evolution. Researchers coming from the field of artificial intelligence such as Kenneth Stanley often see Evolvability as the ability of a system to produce innovative behaviours not observed before [23, 24]. The goal of their research is to exploit the unsupervised process of evolution as an automated optimisation/creation process, that would allow them to find new interesting solutions to difficult problems, without the need for a human designer and unshackled from human cognitive biases. Finally, Richard Watson applies an information theory and theory of learning approach to the concept of evolvability [25]. He defines it as an organism's capacity to encode (e.g. in its genome) adaptive solutions to past selection pressures in a generalisable way, allowing effective responses to novel situations. More evolvable lineages will better extract useful information from past selection events and flexibly apply it to new conditions. Watson, using this perspective, is interested in understanding if an unguided and unconscious process such as evolution can display "cognitive-like" properties similar to learning.

While this plurality of definitions reflects the rich array of ideas that fall under the banner of evolvability, it can also introduce conceptual confusion as sometimes different definitions will point towards aspects of evolvability that are almost incommensurable. For instance, for a researcher who studies evolvability as the amount of genetic variation in a population [26], it can be very difficult to communicate with someone who researches evolvability as the capability to generalise previous selective events and re-use them for future challenges [25]. For this reason, if we want plurality to lead to richness instead of confusion, we need to acknowledge the diversity, and potential incommunicability, between different lines of research both when giving or scrutinising a definition.

The concept of evolvability not only helps expand fundamental theory and research but has relevant practical implications as well. Evolvability can influence applied domains like medicine, agriculture, and engineering, demonstrating the concept's relevance beyond theory. Evolvability could help to predict future events that would threaten human stability such as the rise of new pandemics [27], or the accumulation of antibiotic resistance [28]. On top of being used to prevent risks, research on evolvability could also inform future policies aimed at improving our ecological impact on the planet, such as biodiversity and conservation efforts, containment of invasive species, assessment of the risk and impact of climate change [18]. Moreover, the concept of evolvability could drive research into new advancements to increase our quality of life, for instance, breeding crops for evolvability is a promising approach for tackling complex real-world problems [29, 30]. Evolvability is not only a subject of interest in topics that are related to biological evolution. Evolution has already been employed in the fields of engineering, robotics, and informatics with increasingly impressive results [31–33]. Many researchers in these fields would like to understand more about evolvability in this sense and extrapolate its general properties. Their interest lies in the ability to apply unbounded evolutionary potential for novelty to systems where solutions to real-life problems such as engineering or software development are evolved instead of hand-crafted by humans. This ability of evolution to produce unbounded complexity and novelty is sometimes referred to as open-ended evolution, one of the main research goals especially of the artificial life community [34].

1.3. The View of Standard Evolutionary Theory

Despite a growing consensus, there remains significant debate regarding the importance and prevalence of evolvability in shaping the course of evolution. Standard evolutionary theory has historically paid relatively little attention to this topic and has yet to fully integrate it within its framework [14].

The standard theory of evolution emerged in the early 20th century as a rigorous formalisation of evolution in the language of mathematics and statistics. It led to significant advances in our understanding of evolutionary processes, including the reconciliation of discrete inheritance with continuous traits by Ronald Fischer [35, 36], the concepts of genetic linkage and genetic mapping by J.B.S. Haldane [37, 38], and the metaphor of fitness landscapes by Sewall Wright [39–41]. This work has not only revolutionised the way evolution was understood after its inception. To this day, its principles underpin the fields of population and quantitative genetics, arguably the two most widespread and relevant approaches to the study of evolution.

However, standard theory is not without limitations. At the time of its formulation, scientists lacked the empirical and mechanistic understanding of an organism's cellular, physiological, and developmental processes, and appropriate techniques for studying them. The limited computational power of the time also inhibited the amount of complexity that could be modelled in evolutionary dynamics.

As a result, mathematical and statistical models were developed. These models, by their nature, allow researchers to overlook the intricate mechanics of biological life focusing on observable outcomes. Despite not being fully solvable analytically, equilibrium and stability analyses, as well as approximation techniques can be applied to these models without requiring extensive computational efforts. Studying the dynamics of natural selection, however, is intrinsically complicated and to be able to use these models, limiting assumptions have to be made. For instance, genotype-to-phenotype (G-P) mappings are mainly described as simple 1:1 relations between genotype and phenotype, the effects of genetic mutation are often considered to be unbiased/random and infinitesimal, as the number of loci underlying a trait, and evolution is studied at stable equilibrium conditions with a single fixed selective optimum [36, 42].

This is not to say that standard theory r ejects the possibility of relevant complex interactions between genotype and phenotype, the existence of non-random and non-uniform mutation effects, or the importance of dynamic selective regimes. However, accounting for non-linear interactions, except for particular sub-cases, is extremely hard with just analytical tools, as it is for non-uniform mutation effects or studying evolution outside of equilibrium conditions. Therefore, given these limitations, researchers at the time wisely decided to focus their efforts in directions where the techniques available could be most effectively used to guide research.

1.4. SEEING PAST THE BLIND WATCHMAKER

The focuses and limitations of standard theory limit its scope of inquiry and make it inadequate for the study of evolvability, as they encourage a phenomenological approach to evolutionary study and consequently a gene-centric, short-term view of evolution.

A phenomenological approach aims to describe only the observable outcomes of an event, disregarding the internal mechanisms that lead to it. Classical theory, in this spirit, mainly considers genes (the carriers of information) and phenotypes (the decoded information which is the target of selection) minimising the importance of the internal processes that lead from one to the other.

A gene-centric view of evolution logically ensues from this perspective. Genes independently and directly translate into selectable phenotypes, with no significant intermediate step, and are therefore considered the fundamental unit of selection and the only relevant variable for understanding evolution. Genes are thus seen as selfish entities, in constant competition with one another, coming together with the sole purpose of maximising their own reproduction. It is the selective advantages of individual genes that drive the fate of evolutionary dynamics.

This gene-centric view naturally leads to a “short-term” conception of evolution, eloquently described by Richard Dawkins in his famous analogy of “evolution as a blind watchmaker”. In this metaphor, genes can be seen as the individual pieces of a complex clockwork mechanism (the organism), shaped in myriads of infinitesimally different variations by the effect of small random kinks (mutations), and drawn and assembled by a blind watchmaker (evolution). The watchmaker is blind and has no intention nor foresight, shuffling components randomly with no knowledge of their current role nor future usefulness. Evolution thus can only select for organisms in the present moment, discarding those who fail and continuing to build upon those who survive, unable to select an organism designed to face future challenges; as the watchmaker is blind and therefore oblivious to them.

Evolvability, however, has not been completely dismissed by the standard theory and those aspects that are treatable by a short-term population and quantitative genetics approach have been successfully integrated [15, 26]. This line of research mainly focuses on the role of present variation as raw material for selection and has a fruitful theoretical history : from considering evolvability as a quantity similar to heritability (conceptualised by Houle in his seminal paper [26]), to, more recently, defining it as the alignment of genetic variation with the gradient of selection (the G-Matrix, as done in the on-going work of Hansen [43]).

What these studies do not consider is another crucial aspect of evolvability: the capability to create new variation and to bias it towards adaptive outcomes. Framing evolution this way (phenomenological, gene-centric, short-term, blind) naturally tends to preclude other crucial aspects of evolvability from being a relevant factor to evolutionary dynamics: the capability to create new variation and to bias it towards adaptive outcomes. When organisms cannot be selected for their capability to adapt to new selective challenges and genes constantly fight for their own survival disrupting interactions beneficial in the long-term, it is hard to envision how the ability to produce new variation or to bias it could ever arise.

A broader view of evolvability encompasses both adaptations in response to selection and innovation through generating novelty.

To do so, it is necessary to move from a phenomenological approach toward a mechanistic one [44], where the aim is to explain the nature of the outcomes of evolution in terms of biological processes. This approach is more in line with recent progress and discoveries in the biological sciences. With the omic revolution and the rise of systems biology, it is becoming more and more evident that gene interactions, biased effects of mutations, and environmental conditions play a fundamental role in determining an organism's phenotype [45–47], and theoretical models show that by taking these elements into considerations changes the outcomes of evolution [48–50]. Only when one looks at the inner response mechanism, the innovative potential of the evolutionary process can be understood and explained. Without a knowledge of the inner mechanism, no interesting nor significant explanation can be given as to how organisms generate more or less variation, nor how they can potentially shape this variation in their favour.

In particular, I argue that the design principles and motifs in organismal architecture that facilitate evolvability reveal the otherwise hidden potentials of evolution (e.g. the ability of population to adapt to new challenges not only based on their standing genetic variation, but for instance as well thanks to the ability of the individual's to produce new and possibly adaptively biased variation). By examining the interplay between architecture and evolvability, we gain a richer sense of what evolution can achieve and how it might operate.

Evolutionary developmental biology offers a wealth of examples highlighting the role of organismal architecture in evolvability. The astounding success of arthropods is attributed by many to the modular and flexible architecture of their developmental program, especially of their appendages [51]. The developmental genes of arthropods are organised in such a way that they can be easily combined in different places at different times [52, 53] to produce a wide variety of appendages. This ingenious architecture allowed arthropods to easily evolve appendages to accomplish almost any task: from evolving mouthparts for feeding, to antennae for perception, wings for aerial locomotion, and even gills for breathing [54]. The best instance to display the success of this architecture in enhancing the evolvability of arthropods is the so-called Cambrian explosion. Most of the genes and developmental programs responsible for the burst of diversity in body plans during that period are still being retained to this day [55]. While this variety has been shaved down and fine-tuned to a restricted series of main solutions it still retains the capability for incredible innovative potential, re-exhuming ancestral configurations to face new or returning challenges, as it has happened for instance in the re-convergent evolution of mantis-shrimp raptorial appendages [56].

This example highlights the point that to understand the aspect of evolvability tied to innovative potential, we must turn our attention to organismal architecture and its history.

1.5. A Reminiscent Architect

Understanding evolvability through the lens of organismal architecture can provide insights into the higher-level dynamics that play out over long evolutionary timescales such as those of lineages' successes and downfalls. As a first thing, it is important to define what organismal architecture means for the purposes of this thesis. A general definition of organismal architecture can be given as: "the structural relationship among the elements that compose an organism". What these components are and how they influence evolution changes depending on the scale being analysed, from the genetic to the morphological level. At the genetic level, for instance, architecture refers to the organisation and regulation of gene expression. For example, prokaryotic genetic architecture is often streamlined, with genes linked spatially and functionally, allowing to easily switch on and off entire metabolic functions, while on the other hand, eukaryotic architecture scatters functionally related genes throughout the genome allowing for more interconnected and fine-grained regulation [57]. At the developmental level, architecture describes how different parts of the developmental program interconnect and build upon each other to form morphological structures. For example, it is the architecture of the interactions with other developmental factors of the Hox gene family that allows it to produce many different structures e.g.: limbs, central nervous system, and vertebrae, making it fundamental during development across the tree of life [54]. A similar example can be seen in the expression of Distal-less (Dll) in the formation of arthropod appendages mentioned in the previous section [58]. At the morphological level instead, architecture involves the arrangement and interaction of physiological processes, body parts, and systems that determine an organism's morphology and functioning. The architecture of the chambered heart and lungs in vertebrates varies to accommodate the energetic needs of different clades: the reptilian three-chambered heart allows for partial recharge of anoxic blood, apt for sustaining the low-level metabolism of cold-blooded organisms, while the mammalian four-chambered heart grants maximal efficiency in oxygenation, sustaining high-demand metabolism [59]. These are just a few examples of what the concept of architecture could mean, but as with the concept of evolvability, this term encompasses a wide range of phenomena and its definition cannot be univocal.

The pivotal question this thesis asks is: How can an organismal architecture - how an organism is structured and organised - influence the ability to evolve adaptively?

If natural selection acts in a short-term/blind manner, solely on the basis of an organism's current phenotype and fitness, how can evolvability and adaptive capacity nonetheless emerge over evolutionary time? Together with other scientists [18, 60–62], I argue that this is because the solutions that enable organisms to overcome selective challenges become ingrained in their architecture and that of their lineages. As environments change in the future, this accrued evolutionary baggage can facilitate new adaptations.

This can be exemplified by the increase in the rate of acquisition of antibiotic resistance in bacteria. Bacteria are becoming faster at responding to antibiotic treatments, and more and more bacteria are acquiring resistance to the vast majority of antibiotic treatments [63, 64]. One reason for this lies in their capacity to store past solutions in their genetic architecture. Consider a population of bacteria exposed to a new antibiotic. Mutations that happen to confer resistance often involve the re-purposing of influx/efflux pumps to rid the cell of toxins as well as the modification of the composition of the cell membrane [65]. Those individuals with mutations that apply the right membrane modifications or recruit more of the correct pumps under stress will thus survive and reproduce. If the same bacterial lineage then encounters a similar antibiotic in the future, under similar stress conditions, this genomic memory may aid in evolving resistance more quickly or easily (multi-drug resistance) as often antibiotics belonging to the same or to a closely related family act in similar ways [66].

This example illustrates how the "ghost of selection past" [67, 68], stored in organisms' architecture, could shape evolvability, making it possible for organisms to create and bias variation towards adaptive outcomes. While natural selection alone cannot explain the evolution of features that facilitate evolution itself, the architecture shaped by natural selection gains a form of reminiscent foresight. Over vast timescales, the accumulation of such evolutionary solutions and "memories" in developmental, genetic, cellular, and morphological architectures enable increasing adaptability and resilience. Examples of this are for instance, the plastic tuning of limb length depending on terrain type in anole lizards [69], or the constraint of beak development only in the three relevant dimensions useful to adjust to food type in Darwin's finches [70, 71]. The ensemble of historical adaptations equips the organism with latent abilities to meet new challenges. This view suggests that the dichotomy between "blind"natural selection and the concept of evolvability is false. Dawkins's watchmaker is blind and unguided, mixing genes into combinations, which sometimes by chance, happen to work. But besides it is a reminiscent architect who takes care to integrate and preserve the watchmaker solutions into the organisms' architecture, and uses them as the base upon which to build future adaptations. Thus, the reminiscing architect gradually constructs robust and functional organisms with the help of the blind watchmaker. These organisms contain the record of their past hurdles and can, thanks to this, weather through new conditions. Natural selection forges the structures that enable evolution, broadly conceived, to become an open-ended creative process. Evolvability-enhancing architecture arises inevitably from accumulated experience. The effectiveness of natural selection at solving short-term ever-changing problems thus yields evolvability for the long run.

However, as much as this conjecture might be interesting and the examples brought in its favour compelling, it is extremely difficult to empirically capture the long-term evolutionary dynamics of organismal architecture and its impact on evolvability. Whilst we have sufficient empirical examples that show that organisms possess mechanisms and strategies to create and bias variation toward favourable outcomes [44, 62, 72–76], we understand little of how this mechanism can come to exist in the natural world. Theory can come to our aid and has produced many different models that point towards promising directions for the role of the reminiscing architect in the game of the blind watchmaker.

1.6. INSIGHTS FROM COMPUTATIONAL THEORY

The role of theoretical modelling in the study of evolvability has been paramount since its inception. Being related to long-term dynamics, and involving a plethora of phenomena, evolvability is often difficult to pinpoint in empirical experiments. In particular, this is true for what concerns those studies that analyse the aspects that interest this thesis: the production of adaptive variation in relation to organismal architecture.

To my knowledge, mathematical/computational models addressing the role of architecture, variation production, and its canalisation towards adaptive outcomes, are relatively recent, starting to appear in the literature less than two decades ago. Before that time, the topic of the role of variation and architecture in evolvability was already being discussed (e.g. see the seminal paper from Gunter Wagner and Lee Altenberg [16], the work of Kirschner and Gerhart [19, 62, 76], and way before that the work of Waddington [77]), but the discussion remained on the conceptual level and was mainly backed by verbal arguments. Evolvability was already being modelled, however, this was done from a population genetic approach seeing evolvability as the existent variation in a population [26] and with no regard to organismal architecture (as I already mentioned previously).

In most cases for the models analysing the role of architecture and variation with respect to evolvability, the attention is focused on the regulatory architecture at the genetic level (not to be confused with the structural architecture of genes in the nucleus). These models usually employ various representations of a Gene Regulatory Network (GRN) that leads to the expression of a phenotype. GRN models consider gene-to-gene interactions as the key mechanism from which the phenotype emerges, contrary to the more standard view (population genetics, quantitative genetics, adaptive dynamics, etc.) in which the allelic values are sufficient to derive the phenotype. GRN models often include nonlinear interactions (typically disregarded by standard theory) and model a much higher number of loci than models in population genetics and assume much stronger gene effects than models in quantitative genetics. Overall GRN models encapsulate a more mechanistic approach, which makes them more apt to study the evolution of evolvability and organismal architecture. This type of modelling is not only used to study theoretical evolutionary questions, other disciplines like evolutionary developmental biology (evo-devo) [78] and evolutionary systems biology [79] make as well ample use of these tools and are arguably the fields that first conceived this approach. To my knowledge, a comprehensive review of the different GRN representations in more theoretical/conceptual research is not yet available. Although in no way complete, I will try to provide a brief overview of some of the relevant models and the insights they provided in the coming paragraphs. In particular, I will focus on those models and insights that have guided me to formulate my conception of the relation between evolvability and organismal architecture. These insights are:

- The capability of architecture to evolve through natural selection to bias mutations towards adaptive outcomes.
- The fact that complex architectures can easily evolve to robustness: i.e. reduce the

negative effects of deleterious mutations and enlarge the search space of evolutionary solutions.

- The proof that architectures in changing environments can evolve to become modular i.e.: having an independent functional component that can be flexibly changed and recombined
- And finally but most importantly that architectures indeed have a form of "developmental memory" or as I call it reminiscent foresight: i.e. they can retain solutions to the past selective challenge and redeploy them when in need.

Some of the first models to explicitly address the relationship between evolvability and architecture, are those from Pauline Hogeweg [59, 61, 72]. Hogeweg and collaborators show how natural selection can spontaneously restructure architecture to lead to the production of adaptively biased variation. They model an organism's GRN with a complex and evolvable architecture representing its metabolic pathways. GRNs evolve in a regime of two changing environments (i.e. selective challenges) that require switching between two different types of metabolism. Individuals in these simulations evolve evolvability through simple natural selection, becoming increasingly faster at adapting to changes in the environment. This is made possible as the architecture of the GRNs evolves to minimise the number of mutations needed to switch from one optimal phenotype to the other and reduce the number of deleterious mutations that could afflict the network. Some architectures adapt so well that they are able to evolve a "switch" structure, that with a single mutation makes the organism adapt from one environment to the other.

The models of Andreas Wagner address the influence of architecture on the effects of mutation and variation as well. A good part of their work focuses on a specific architectural property: robustness [80–83]. Robustness is the capability of a system (i.e. an organism) to undergo multiple perturbations (i.e. mutations) without disrupting its overall function (i.e. changing the phenotype) [84]. A. Wagner and his collaborators show in a series of models of both GRNs [80] and RNA sequences [83] that robustness is a property that can easily evolve in these systems and that, as already claimed by other researchers in systems biology [84–87], robustness favours evolvability. This at first glance might seem counterintuitive, since weaker effects of mutations should lead to less phenotypic variation and innovation, thus reducing evolvability. However, A. Wagner and collaborators demonstrate that this is a false dichotomy. Robust systems allow the evolutionary process to travel along "pathways of neutral mutations" (consecutive mutations in the genotype that do not perturb the phenotype); joining adaptive phenotypes that would otherwise seem impossible to reach through "blind" natural selection, for instance, because the most direct path between the two is made of maladaptive steps. Being able to circumvent maladaptive pitfalls to reach more adaptive phenotypes provides robust architectures with a higher "heritable phenotypic variation", making them more evolvable [83]. A. Wagner imputes robustness to how the architecture influences the mapping from genotype to phenotype (G-P map). In particular, he advocates that it is the disparity in the degrees of freedom/dimensionality between genotype and phenotype spaces that makes this possible. When modelling genotype-to-phenotype translation as a network of interactions (i.e. mechanistically), the number of possible connections and combinations between its elements is exponentially higher than the number of possible outcomes (in a similar way to how the number of possible neural connections in the brain is much bigger than the atoms in the universe, while the possible number of behaviours a brain can express arguably is not). This naturally leads multiple genotypes to map onto the same phenotype, thus making it easier to find neutral phenotypic paths that can lead to adaptation. The idea of the role of high-dimensionality and redundancy in G-P landscapes playing a role in evolution is reprised by other researchers, see for instance Gavrilets 2010 [88] and [50].

Another important line of research that focuses on the effects of architecture on the G-P landscape (like Andreas Wagner) and variation bias in changing environments (like Hogeweg) is that of Günter P. Wagner and Jeremy Draghi [89, 90]. Together with Lee Altenberg, G.P. Wagner is the proposer of the first “variability-based” definition of evolvability: “Evolvability as the ability of random variations to sometimes produce improvement.” [16]. In the same article, he also advocates the importance of another “architectural” concept for studying evolvability: modularity [70]. Modularity is the property of a system where functionally distinct parts work independently from each other [70]. In an organism with a modular architecture, different components will perform different functions (i.e. during development different sets of genes will take care of developing different sets of organs), making it so that mutation on one component won’t hinder the functioning of another. Like robustness, modularity is a design principle that has been long recognised as important in organismal architecture and evolvability. Modularity allows evolution to freely re-shape one component to a new task without compromising the functionality of the whole organism, thus increasing the range of possible solutions to a selective challenge [16, 86, 87, 91]. G.P. Wagner and Draghi show in a modelling study both that architectures evolving in changing environments can imprint onto themselves the different selective challenges they face, organising different adaptations into separate functional modules [89], and that architectures can bias variation toward adaptive outcomes, strategically positioning themselves in the G-P space so that few mutations can switch from one to another adaptive phenotype if the environment changes [90].

G.P. Wagner has contributed to a wealth of theoretical studies on evolvability and architecture, often collaborating with researchers from other disciplines that are more computationally oriented. One collaborator was Richard Watson [60], a researcher with a background in computational and cognitive sciences. Watson and collaborators demonstrate with a GRN model how evolution can display “learning-like” behaviour. They evolve GRNs to display target expression patterns. Once the GRNs are perfectly adapted, researchers scramble the network’s configuration to become completely random and show that nonetheless, if the architecture is preserved, said GRNs are able to “remember” and eventually re-express the expression pattern they were selected for. Even more surprisingly, if the GRNs are evolved to display not only one but various target patterns, once disturbed they not only can reproduce one of the previous patterns but are able to produce patterns that are the synthesis of the ones they adapted to. This indicates that evolution is not only able to remember but to generalise over past adaptations. Watson, together with Eörs Szathmáry, articulates the vision and possible implications of comparing evolutionary processes and their relation to evolvability to learning processes in a subsequent paper [25]. Here they state:” evolvability is to evolution as generalisation is to learning” and I would add, as they implicitly do in the paper that “Organismal architecture is to evolvability as memory is to generalisation". Indeed what this, as well as subsequent research from Watson shows, the place in which "developmental memories" are stored and engrained, is none other than the structural relations between the components of their GRNs [92].

The conceptual framework that Watson applies to his research and the way he frames his insights are not native to evolutionary theory. Most of the concepts he adopts originate from the field of computational science. In particular, the terminology Watson uses when discussing evolution as a learning process is rooted in the concept of learning or training in Artificial Intelligence (see [25, 92]). This is not exclusive to Watson's research, in the line of research on evolvability and architecture there is no lack of crossovers from other disciplines. For instance, many papers on the evolution of modularity often model the evolution of circuitry or robot controllers, clearly showing their roots in engineering disciplines that stepped over into evolutionary theory [91, 93, 94]. This is not a coincidence as the evolutionary, engineering, and computational fields share many points of interest, the most evident one being: the adaptation of complex systems to complex tasks [59, 95]. The use of evolution is indeed no stranger to many lines of research that have nothing to do with biology, and are instead interested in solving more practical problems such as optimal circuit configuration, algorithm design, and artificial intelligence [96, 97]. Most of these practical problems, in one form or the other, are often related to some form of computation, and indeed the field of Evolutionary computation has been existing now for almost half a century [97]. As the example from Watson clearly shows, integrating decades-long research on evolution and evolvability from other disciplines could prove to be very fruitful for evolutionary biology, and the opposite is also true. For this reason, I would also like to give a quick overview of studies outside biology that also focus on architecture and evolvability.

1.7. CROSSOVER: EVOLVABILITY AND ARCHITECTURE IN COMPUTATIONAL SCIENCES AND OTHER DISCIPLINES

A first example is that of research conducted by Jeff Clune (who also collaborated with Richard Watson in) together with Jean Baptiste Mouret and Hod Lipson [98]. All these scientists come from purely computational or robotic engineering backgrounds, nonetheless, their research had important resonance also in the biological community. In their paper, Clune and collaborators show that modularity can evolve in networks in very simple conditions, namely: by simply imposing a cost on connections. They evolve networks similar to Artificial Neural Networks (widely employed in Artificial Intelligence) and select for their proficiency in pattern recognition and Boolean logic tasks that change over time. They demonstrate that networks that pay a fitness cost for their connections, evolve to minimise this cost and become more modular, while those networks that are not subjected to connection costs, never evolve a modular structure (they remain integrated [91]). Moreover, and more importantly, they go on to show that modular networks are more evolvable, being able to adapt to new tasks faster and better than those that remain integrated. Despite this paper originating from researchers whose backgrounds are only tangentially related to biology, the similitude is obvious between this system and how we think modularity could favour evolvability in organisms [70, 99–103]. A similar study was conducted by Kashan a few years earlier on the evolution of logic-gate circuits [93] (the building blocks of computational hardware).

Another interesting research line is that of Kriegman [94] in the field of evolutionary robotics [104, 105]. In this discipline, researchers use evolution to find new and innovative robot designs or robot controllers. This often involves a form of development, where the body of the robot and/or its controller are not directly implemented, but instead emerge dynamically from the instruction of a genetically encoded developmental program. For example, Kriegman and colleagues evolved soft-bodied robots for a locomotory task [94]. The development of the body design as well as the locomotory controller of the robot are divided into two separate modules. This decoupling increases evolvability, through a process called differential canalisation. During evolution, the two developmental systems take two different evolutionary pathways: the development of the body canalises [77], becoming robust to mutations, and reliably presenting a working body design independently of the controller (modular behaviour). This, in turn, allows evolution to experiment with a range of controller solutions without the risk of disrupting the overall ability for locomotion. This way the overall evolvability of this system is increased, as it can explore more solutions through the controller, and at the same time it buffers the effects of maladaptive variations with a robust body design. Kriegman and collaborators speculate that this pattern of differential canalisation could be driven by the evolutionary advantage of having a high bias towards neutral or adaptive mutations (for the controller) at the cost of losing variability (canalisation) in the function of another component (the body design). Finally, they relate this concept to the idea of genetic assimilation by Marie Jane West-Eberhard [106] (the constitutive expression of the previously plastic expression of a trait) and propose it as a possible explanation.

Both examples display how insights and approaches from different disciplines can enrich our understanding of biological evolution, helping us to illuminate the role of architectural principles in guiding evolvability. Nonetheless, the two pieces of research clearly nod in the direction of evolutionary biology, and their goals remain largely theoretical and aligned with those of a good portion of evolutionary biologists. There are however branches of research in computational sciences, that specifically focus on using evolutionary principles to resolve very practical problems, and have very little interest in the biological coherence of the underlying concepts. These fields are for instance Neuroevolution (evolution of the architecture of Artificial Neural Networks) [107], AI evolutionary algorithms (the design of different modes of evolution, normally for evolving/training such networks) [31], and genetic programming (the evolution of programs) [108]. What is a fascinating aspect of these disciplines is that the scientists involved often approach evolution from a completely different perspective than a biologist would. Their final goal is to evolve either an artificial neural network, an algorithm, or a program that is the best at executing a certain task. Little interest is given to the soundness of the rationale behind design choices. Instead, the value of a result is usually established based on clear and practical benchmarking tests, which often involve outperforming or matching the quality of results of other state-of-the-art methods. Therefore, researchers in this field do not try to reproduce evolution as it is in the biological world. They, instead, retro-engineer it, taking inspiration from biology only when useful, and try to create the optimal design for evolution to solve a problem of interest (i.e. playing Atari games, robot locomotion, controlling virtual football players [109]). As scientifically flawed as it might sound to a biologist, this more unrestricted way of studying evolution (not biological evolution, just evolution as an algorithm) has led to many interesting insights and creative evolutionary outcomes [110]. I argue that achievements from these foreign disciplines can serve as an inspiration to biology [111] and advance its line of research (of course the opposite is also true, see Ch 4 of this thesis).

Of particular interest to me is the fact that despite their detachment from biological principles and evolution, we find that architecture and evolvability are still central ideas in this research.

Research from Kenneth Stanley and Risto Miikkulainen proves that an algorithm that allows neural networks to evolve by gradually augmenting their topologies (NEAT) can outperform fixed-topology trained networks in many tasks [112]. The intention behind NEAT is to allow evolution to “both optimise and complexify solutions simultaneously, offering the possibility of evolving increasingly complex solutions over generations” [112]. NEAT does so by maintaining and incrementally improving via evolution parts of the network architecture. The idea of evolvability (defined by Stanley as the ability to produce innovation) is also very involved in the conception of NEAT, and indeed in many cases, NEAT has been proven to enhance the evolvability of the system in which it is used [113, 114]. NEAT has given rise to a very prolific line of research and is one of the best-known representations of ANN evolution [94, 115–119]. Its framework has now been expanded for almost two decades in a family of architecturally oriented encodings that allow a more efficient evolution of artificial neural networks for a plethora of tasks.

Another interesting architectural insight provided by Kenneth Stanley and Sebastian Risi is their proof that evolution can form very complex and high-performing ensembles by being able to find a way to efficiently connect high-dimension components (in their case architectural blocks of artificial neural networks with hundreds of thousands of connections/parameters each) through low-dimension interfaces (i.e. through only a few connections) [120]. This process mirrors the fashion in which different developmental programs interact only via a few of the many genes that compose them [70]. Even more interestingly, they show that mutations on these “compressed” low-dimensional interfaces produce coordinated and functional changes across the “expanded” high-dimensional components. This behaviour is well-known in artificial intelligence and is specifically exploited by the “Auto-encoder” architectures [121, 122] (sometimes referred to as hourglass networks). Autoencoders are famous for being able to produce generative art and the (sometimes hilarious, sometimes scary) so-called deep-fakes [123, 124]. They can do so via compressing and expanding information, thus forming information bottlenecks. Scientists have shown that these bottlenecks contain the crucial features of the incoming information and are thus much easier to fine-tune for a specific task [121, 122]. This leads to speculation that biological organisms might be able to bias variation adaptively in a similar way. Bowtie architectures are well-known across biology and very similar to auto-encoders. They have the same low-dimensional information bottlenecks that connect different high-dimensional components [125, 126]. Mutating preferentially these variational hotspots at the interface of different functional systems thus, like autoencoders, could favour the production of coordinated and functional variation. This idea is not completely new in evolutionary biology and it has, in different terms, already been presented by Kirschner and Gerhart in their theory of facilitated variation [62]. I suspect, however, that Kirschner and Gerhart were oblivious to this "convergence" with Artificial Intelligence, as auto-encoders became popular only in the recent decade [124]. No one to my knowledge has ever explicitly made the connection between these two ideas until now. However, it has to be said that autoencoders are being used in evolutionary computation already and that Richard Watson as well has been recently working with this representation [127].

Lastly, I would like to focus on Grammatically Guided Genetic Programming (GGGP) [128], a branch of genetic programming (GP) [108]. The goal of GP in general, as already mentioned previously, can be outlined as the attempt to design programs to solve specific computational tasks via evolution. This is thought to be a particularly fruitful approach for those computational tasks to which no known optimal solution exists (i.e. calculus optimisation, path-making algorithms, robotic arm kinetic control, ANN training optimisation), and where therefore evolution's creative potential can shine. GP represents programs as binary trees, which can be easily parsed recursively and executed, and the structure of these trees is genetically encoded in a variety of ways. Practitioners of GGGP explicitly design all the different operators of the program and encode them into a grammar system. In this grammar different types of operators cover different grammatical functions (e.g. a sum operator conjoins two other operators, a parenthesis operator forms a new branching point in the grammatical sentence, a numerical constant closes the sentence, etc.). Although this is referred to as 'grammar', the base idea of this line of research is to use expert knowledge about a problem to design a system of relationships between operators so that evolution can more easily explore the solution space i.e.: to design the architecture of a system to increase its capacity to adapt - evolvability. This approach has been successfully applied to a variety of tasks, from designing neural architectures [129] to image recognition [130], and pathfinding [131]. Its explicit attention to architecture, as well as its general applicability [132], make GGGP a fertile ground for applying and gleaning insights on architecture and evolvability (see Chapter 4 of this thesis).

1.8. METHODOLOGY

Now that I have stated my ideas and provided sufficient conceptual background, I can illustrate how I intend to research the role of organismal architecture in evolvability and the rationale behind my choices. A rationale is indeed necessary for this line of research, since, as I hope to have sufficiently demonstrated in the previous sections, one defining aspect of evolvability research is the astounding plurality of all its aspects, from the conceptual (discussed above) to the methodological (discussed below). The examples given in the previous two sections provide a good survey of the diversity of techniques and approaches one could apply to the study of architecture. From different choices of representation of architecture e.g.:

1. a GRN described by interconnectable metabolic reaction rates,

2. a GRN that is a fully connected binary network,

3. a GRN that resembles a simple ANN,
4. a GRN that mimics a logic gate circuit,
5. a complex state-of-the-art Artificial Intelligence Neural Network Architecture,
6. a robot morphological and motor developmental program,
7. a grammatical system (see GGGP)

to different choices in the set of selective challenges in which individuals evolve e.g.:

1. Optimal metabolic end-product expression via equation system optimisation,
2. Stable matching of one or multiple expression patterns,
3. Execution of alternating Boolean logic operations,
4. Pattern recognition of modularly changing bit sequences,
5. Simultaneous and sequential binary output matching,
6. Emergent speed of locomotion,
7. Image recognition, path-finding, or robotic arm coordination

Modelling choices are almost limitless, and this introduction does not even get close to covering all of them (for other systems in use see RNA landscapes used by Hogeweg and Wagner [59, 83], or Artificial life systems such as AVIDA [20], Tierra [133] and the various iterations of the cellular automata models [134]). On top of this, I would like to mention that I did also not cover in detail the role that different selective challenges play as well in influencing evolvability (as this exceeds the already hefty scope of this introduction).

In the face of so many choices, one can easily feel overwhelmed. Moreover, one needs to recognise that, despite some of these models do try, and in some rare cases even succeed, to accurately capture some aspects of biological reality, there is a huge gap, both qualitative and quantitative (see Discussion), between the outcomes predicted by theoretical models and the natural world. Therefore, it is important when making a model to face the immense sea of possible implementations with clear goals and intentions, as well as to acknowledge the difference between models and reality. To do so, I choose an approach of intermediate complexity (in line with the philosophy of my research group). The idea of intermediate complexity is enucleated in the eloquent (yet so devilishly vague!) quote from Albert Einstein that says: “Make everything as simple as possible, but not simpler.” [135] which I would rephrase, to help the reader understand, as: “Keep everything as simple as possible, then add one grain of complexity, and then, again, keep this grain as simple as possible.”. The idea of such an approach is to start from simple yet reasonable assumptions, understand where the limits of those assumptions are, and in a methodical and almost naively simple manner, relax one assumption. One should strive to make the most simple and yet most significant change possible, and then let the new properties and outcomes of the so-changed process spontaneously emerge. Once this is done, and ideally all ramifications have been assessed, the process can be iterated again. Intermediate complexity is a “what if” approach, that seeks not to predict real events, but to give us a glimpse of what other potential dynamics we might be missing out on by restricting ourselves to a certain set of assumptions. This is the idea behind my research methods: through models, not to describe and predict what the interplay between organismal architecture and evolvability actually does, but to prime our intuition (mine and that of every other scientist interested in the topic) to what it could be doing. To ground this idea more concretely, I will now illustrate my modelling choices in researching evolvability and organismal architecture. Let me first recapitulate the principles and limitations of standard evolutionary theory. This theory finds its strengths in its simplicity and overall cohesiveness. Evolutionary processes are described in a mathematical and statistical framework, where individuals are often subsumed into populations and specific traits to average quantities. This allows it to describe a wide range of complex dynamics in simple terms, and, accepting the simplifications that come with it, to rigorously predict their outcomes. However, as I already explained, this approach also leads to a phenomenological and short-term view of evolution. Under this view, only individual traits and not their interactions are selected, and long-term evolutionary advantages cannot be accounted for by natural selection. For the theory to move on from this impasse, and be able to account for phenomena like evolvability, it is necessary to take a mechanistic approach, acknowledging in particular that organismal architecture can provide new and important insights. Therefore I aim, in line with my rationale, to implement the simplest expansion possible to be able to model systems so that variation between individuals is explicitly accounted for, and interactions between genes and their architectures are relevant to evolution. This way I hope to observe in minimal conditions if and how evolvability emerges from their newly expanded dynamics. For this reason, my simulations have two key expansions from classical theory: each individual is explicitly modelled (instead of being averaged at the population level) and its phenotype is determined by the outcome of the activity of a GRN with a minimally complex architecture (instead of separate genes with little interaction among each other). A good approximation of the modelling method I use can be seen in the research from G. Wagner and Draghi [90], as well as van Gestel and Weissing (specifically the GRN) [50]. Individuals’ GRNs are usually composed of a few, but tightly interconnected, genes. The GRNs have (innately) an architecture of connections, but this usually has an extremely simple topology that cannot evolve. Overall, their functioning and behaviour are very similar to some of the most elementary ANN architectures i.e.: multilayer perceptrons [136, 137]. This similarity is not a coincidence, as it has been mathematically proven that these basic ANN architectural templates have all the minimal requirements (if scaled to a sufficient size i.e.: number of genes/neurons) to be able to accomplish any given task (i.e. approximate any given function/ be Turing complete) [138]. This makes them ideal for my goals as they are the simplest possible mechanistic representation of a gene network that still allows for new (and potentially any) behaviour to evolve. These choices, however, exclude the observation of many interesting dynamics, such as the evolution of modularity, particular design motifs, or highly complex behaviours. At the same time, this allows me to focus my analysis on other aspects more clearly (in a similar way to how the standard theory foregoes most organismal complexity in exchange for exceptional clarity in its formalism). By maintaining such a low degree of complexity, it is possible to capture dynamics in systems so simple that they are easy and intuitive to understand for a wide variety of scholars, making this concept more accessible to the entire community. Moreover, simulations, in some cases, are simple enough that they can be translated from a computational model into a mathematical formula. This re-formalization could help to bridge the gap between the phenomenological/mathematical approach of the standard theory with the mechanistic/computational approach that evolvability studies normally adopt. In conclusion, I hope to highlight the advantages this approach offers, and why I find it particularly appealing in trying to "pick up" a general pattern across many phenomena and combine knowledge from different fields. Overall my modelling does not want to produce digital experiments, but instead aims to provide computational metaphors; continuing Waddington's conviction that models should be "Tools for Thought" [139].

1.9. THIS THESIS

This final section will give a brief overview of the contents of this thesis. The work in this thesis will first deal with conceptual work on evolvability (Chapter 2), it will move on modelling (Chapter 3 and 4 and the two intermezzos), to then give an example of multidisciplinary research between evolutionary fields (Chapter 5), and finally will conclude with a few philosophical consideration on the field of evolutionary studies and a distillation of this thesis' results (Chapter 6 / Discussion).

Chapter 2 discusses the many facets of 'evolvability' and argues that a mechanistic perspective can clarify and resolve long-standing debates in the field and facilitate communication between different disciplines. Thinking in terms of mechanisms can help to more clearly identify what facet of evolvability one is studying, allowing one to more easily pinpoint the scope and limitations of one's research. We also review the current definitions used for evolvability and highlight a previously unremarked difference in the way different groups of researchers conceptualise it.

Chapter 3, provides a 'proof of principle' that adaptive developmental biases at the phenotypic level can readily evolve, even if the mutation process at the genetic level is unbiased and random. We term the particular type of mutational bias we are interested in: "mutational transformation". A mutational transformer is a mechanism through which the phenotype switches from one adaptive state to another via a single or few genetic mutations. Using individual-based simulations, we show how mutational switches can evolve in gene regulatory networks with simple and complicated architecture. We observe two distinct modes of mutational switching depending on the type of allowed gen interactions: mutational amplifiers for linear interactions and mutational canalisers for nonlinear.

My thesis also includes two 'intermezzos': research that I conducted in the initial stages of my PhD. Both intermezzos were published as part of a larger study. I here only reproduce the theoretical aspects of the study, as they are relevant to Chapter 4 of this thesis.

In the **first intermezzo**, which is part of the published paper. I discuss another potential application of the mechanistic modelling approach to study the evolution and evolutionary consequences of non-genetic inheritance. I first give an overview of the many different models with which evolutionary biology has described and treated non-genetic inheritance, highlighting their strengths and pitfalls. I then proceed to describe a potential model in line with my methodology, which would allow the incorporation of the molecular and evolutionary insights collected in the publication.

The **second Intermezzo** includes a modelling study part of the published paper [140] that provides an explanation for the puzzling fact that parasitic wasps can be induced to switch on their fat metabolism, although they seemed to have lost the corresponding metabolic pathway millions of years ago. This finding seems to contradict Dollo's law of irreversibility, which states that complex traits cannot be re-evolved once lost. The model demonstrates that a switching device (switching on or off a metabolic pathway) can be stably maintained in the face of mutational erosion, even if the switch is only rarely used. This supports that the wasps' metabolic pathway persists plastically dormant rather than re-evolving after irreversible loss.

Chapter 4 extends the study in Intermezzo 2 by considering the evolutionary emergence and robustness of gene-regulatory networks with a complex task (not just a switching device) in situations where the adaptation is only sporadically under selection. The model reveals complex reaction norm phenotypes evolve under regimes of sporadic selection. Both genotypes pre-adapted under constant selection, and those not pre-adapted perform equally well. Counterintuitively, larger genotypes withstand mutations better in certain conditions. Overall, the results demonstrate complex regulation can emerge even with minimal selective pressure.

Research in **Chapter 5** leverages recent insights from evolutionary theory to improve the design of Genetic Programming (GP) algorithms aimed at Machine Learning. Specifically, mutational regimes are designed that improve the search efficiency and performance of a certain class of Neural Network optimisers. The new mutational regimes prove to be computationally efficient while at the same time increasing the quality and diversity of an image recognition task, demonstrating cross-fertilisation between fields.

Finally, **Chapter 6** contains an overarching discussion of my thesis. This chapter contains two parts. First, I lay out some considerations and conclusions of a broad scientific/theoretical and philosophical nature. Second, I review and summarize the findings of my research and outline interesting future directions for this field of study.

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6

CONCLUSION

There will be time, there will be time
To prepare a face to meet the faces that you meet;
There will be time to murder and create,
And time for all the works and days of hands
That lift and drop a question on your plate;
Time for you and time for me,
And time yet for a hundred indecisions,
And for a hundred visions and revisions
Before the taking of a toast and tea.
...
Do I dare
Disturb the universe?
In a minute there is time
For decisions and revisions which a minute will reverse.

TS Eliot, The Love Song of J. Alfred Prufrock

In this final chapter of my thesis, I will try to convey the scientific insights I have gleaned during my trajectory as a PhD candidate. However, first, I think it is important to acknowledge that research is conducted by people (me in this case) and that people, willing or not, have personal and biased opinions. This chapter is written with a philosophical view in mind. To promote a clearer understanding of my conclusions, or at least a clearer perspective on the importance I attribute to them, I think it is necessary to articulate this view. To this end, I will divide this chapter into two parts: In the first part, I will try my hand at some philosophical considerations, to provide clarity about what my perspective is. In the second part, I will summarise my findings and the conclusions I draw, as well as the limitations and future directions I see for this line of research.

6.1. PART 1: A POST-MODERN SYNTHESIS

6.1.1. A POST-MODERN PUN

The section title “A Post-Modern Synthesis”, is obviously a pun, playing on the name of one of the bulwarks of the evolutionary theory: the Modern Synthesis; and the now so popular, although not universally liked, philosophical term: post-modern. The aim of the pun is to show what my (somehow post-modern) philosophical standpoint is concerning the study of evolution. I think that we should abandon some modernist categorical concepts of evolutionary thinking (which I may unjustly associate with the Modern Synthesis here) in favour of a more supple, more post-modern, demeanour. The Modern Synthesis of Evolutionary Biology refers to an attempt at a comprehensive explanation of how evolution works developed in the early 20th century. It tries to reconcile Darwin’s theory of natural selection with the principles of a population-oriented view of Mendelian genetics emphasising that: (1) Mendelian genetics can be reconciled with quantitative genetics (at that time called Biometry [1, 2]), and (2) the laws underlying both mendelian genetics and quantitative genetics are both compatible with evolution by natural selection (i.e. improved adaptation over the generations). The latter is not self-evident, as it had been shown by Weismann that Darwin’s own theory of inheritance was incompatible with his ideas on adaptive evolution (see [3] for an in-depth rundown). On its ideas, the standard evolutionary theory was constructed, with all its advantages and downfalls (phenomenological, short-term, gene-centric; as already discussed in **Chapter 1**). The standard theory has been helming research for almost a century now. Nonetheless, in recent times some evolutionary biologists, counting me amongst their ranks, say that the standard theory is too restrictive to paint the full picture. Many have called for expansions to the paradigm, but it seems that it is extremely difficult to come up with a general evolutionary “synthesis”. And here enters the post-modern part of this pun. I argue that if we are ever going to have a synthesis in Evolutionary biology this is going to be a post-modern one, a pluralistic one. Post-modernism arose from the rubble of the Second World War which was, for many, the end of the Modern Era. It ushered in a cohort of ideas that made Modernity tremble, such as there is no universal truth, progress, or good, matter comes before meaning, and human language is inadequate to communicate beyond our basic needs. Amid this philosophical storm of “slings and arrows", nonetheless sat a smouldering ember of hope. A hope that decided not to ignore or justify our incomprehensible and contradictory nature, but to embrace it in all its facets. A hope which still dared to say: there is something worth it, but it is not that easy! This is what I see in post-modernism, and this is why I find myself adhering to some of its views. It is from the philosophy of post-modernism that I take some ideas I would like to apply to evolutionary biology, in particular, that of pluralism.

6.1.2. PLURALISM IN POST-MODERNISM

The idea of pluralism is best embodied, in my opinion, by Simone de Beauvoir in: "The Ethics of Ambiguity". The message from Beauvoir's work is that we should do away with a logic made of dichotomies. Things are neither black nor white, instead, they often are points in the middle of one, or even many, spectrums. One can only hope to see the world as completely as possible by looking through as many of its facets as possible. It is important, however, to note that pluralism should not be confused with relativism. Pluralism does not imply abandoning basic rules of argumentation: arguments need to be logically correct, and sound and can, and should, be evaluated against each other. However, many arguments can coexist to explain the same phenomenon. In sum, I use post-modern in my title to refer to pluralism as one of the ways to obtain an actual evolutionary "synthesis". Advocating for the idea that in evolutionary studies, and science in general, many views should not entail conflict per se, but instead, they allow us to appreciate evolution in a richer way. Pluralism does not only entail different views. You could also have pluralism in terms of explanation or methods. Different approaches can suit different purposes: e.g. population genetics to predict the changes in frequency of beneficial alleles in the upcoming generations of artificially selected crops, and evo -devo to discover new evolutionary mechanisms these alleles allow.

6.1.3. PLURALISM IN EVOLUTIONARY BIOLOGY

I argue that this pluralistic view has its uses in evolutionary biology (and science in general) and that it can help us advance our understanding of the natural world. The relation of pluralism to scientific research is metaphorically similar to the relation of a multidimensional fitness landscape to the study of adaptive trajectories [4] or the relation of a complex Gene Regulatory Network (GRN) to the study of evolvability and robustness [5] (see **Chapter 1**). By adding dimensions, and therefore degrees of freedom, to evolutionary models (e.g. looking at fitness landscapes with more than three dimensions [4], or considering more realistic, and therefore more intricate, Genotype-Phenotype maps [5]) evolution has many more paths to achieve a certain adaptation than previously expected. In the same fashion if we, through pluralism, enlarge the dimensions in which we can approach evolutionary questions via different perspectives (e.g. via a population genetics approach, a mechanistic GRN approach, or seeing evolution as a learning process) we can increase the number of research lines we can leverage to explain evolutionary phenomena.

Some lines of research not only serve as a metaphor for the usefulness of pluralism but are in and of themselves an example of this pluralism in action. What I mean by pluralism in this case, is the willingness of certain scientists to detach themselves from the traditional paradigm with which some natural phenomena are interpreted, and approach them from a completely different angle. In a certain sense, I equate pluralism to a very strong form of multi-disciplinarity, where not only findings from one field can be used to enhance research in another (see for instance the recent improvements in protein-folding prediction thanks to artificial intelligence), but the entire frame of reference is used. This attitude in my opinion can lead to significant leaps forward in our scientific understanding with relatively little effort.

I will quickly discuss my interpretation of the three examples already mentioned in this thesis: the Modern Synthesis, the work concept of bioinformatics, and the interpretation of evolutionary processes as a form of learning. The Modern Synthesis is perhaps the most famous and important example of pluralism applied to evolutionary studies. It is common knowledge that Darwin left this world with one regret: his inability to explain the mechanism of inheritance. He knew that this missing explanation could undermine its entire theory. It was the work of Gregor Mendel and his discovery of the gene that solved this conundrum (alas! a bit too late for Darwin). However, initially, Mendel's work, by proving the existence of discrete units of inheritance (the genes), did seem to put Darwin's theory in jeopardy as it could not explain the presence of continuous traits in nature. The work of Fisher, Haldane, Wright, Huxley, and many other scientists who contributed to the Modern Synthesis at the time allowed us to reconcile this seemingly fatal flaw of evolutionary theory and push it forward. They did so by bringing into the fold of this discipline the language and ideas of statistics and physics. This allowed them to explain the complexity and continuity of the natural world through its discrete components. If not for the work of these scientists, and the research that followed from it, the research I conduct on evolvability would most likely not exist. Another important example that also played a role in my research is that of bioinformatics by Pauline Hogeweg. As Hogeweg explains in detail in her opinion paper [6], she and Hesper started using this term in the 1970s. By choosing the name bioinformatics, they starkly outlined their research against other fields such as biochemistry or biophysics, making it clear and immediate that t he focus of their research was the way information is transmitted and transformed, and that they considered this aspect a key feature of life. By explicitly looking at how information flows in biological systems, their intent was to bridge the gap between phenomena at different organisational scales via emergent properties. This information-focused computational approach has arguably given birth to a whole branch of more mechanistic, individual-based, and emergent models. These types of models would later be adopted by evolutionary biologists (such as Gunther Wagner, Andreas Wagner, and Hogeweg herself) to study aspects of evolution normally inaccessible to standard theory. Most of the research on evolvability comes from this branch, and it is in its spirit that I conduct mine as well.

Finally, also connected to ideas from informatics, the work of Richard Watson is, as well, an excellent example of a pluralistic approach in evolutionary biology. Richard Watson, a computational scientist by formation, takes ideas from the field of Machine Learning and applies them to interpret evolutionary processes. In two seminal papers [7, 8] Watson traces the interesting parallel between evolution and reinforcement learning, one form of learning in Artificial Intelligence, used to solve complex problems for which the solution is unknown. By unveiling this similitude, Watson brings a new perspective to explain evolvability, where each past evolutionary success is imprinted and stored in the organism's "memory". A process that gradually leads to the "generalisation" of a set of selective problems, allowing organisms to deploy/evolve efficient solutions more and more easily. It is in the light of these insights that I developed my ideas and defined my focus in the vast field of evolvability research. I did this though with the additional insight, coming from Hogeweg's research, that it is in the representation (i.e. the architecture) that one can observe how this process takes place.

In conclusion, pluralism in these different approaches does not only benefit the individual research lines. Evolutionary biology as a whole can, in the light of pluralism, benefit from their coexistence. The study of evolution is a young science when compared to physics, chemistry, or even natural biology, with just a little over a century of research to its credit. Yet evolutionary biology poses some of the most complex questions about the most complex forms of matter in the known universe: life. No star nor galaxy holds a candle in terms of complexity to the metabolism of even the smallest bacterium (its biology), even less so to its interactions with other species (its ecology), not to mention its changes through time (its evolution). For this reason, we are far from having clear answers to most of these questions, and sometimes there is even much debate on their validity in the first place. A pluralistic approach, with its richness of views and interpretations, can help us chart this unknown territory that is evolution. In this endeavour, different types of research can occupy different niches, pushing forward the overall progress. Scientists such as Watson and Hogeweg propose new and explorative theoretical concepts, others work to incorporate these insights into existing frameworks and generalise them, and others still strive to apply them in real-world contexts. In this pluralistic "research ecosystem", I would like my niche to be one that stimulates research by creating a new lexicon of evolutionary dynamics. Via the use of conceptual models that strive to push the boundaries of current theory, I aim not to describe or predict the natural world directly, but instead to tease the attention of the community towards new interesting phenomena.

6.2. PART 2: INSIGHTS ON EVOLVABILITY AND ARCHITECTURE

The body of my insights can be divided into two types: those I gained from reasoning and researching broad topics related to evolvability aiming to combine and connect ideas in new ways (Chapters 1, 2, and Intermezzo 1), and those which I derived from analysing computational models, often involving a mechanistic comprehension of the role of architecture in evolvability (Chapters 3, 4, 5, and Intermezzo 2). When talking about the "architecture" of a GRN in my models, I typically refer not to different topologies (number of nodes and the way the nodes are connected), but to the specific configuration of the strengths of the gene-to-gene interactions. I also conducted some research on evolving GRN topology together with a Master's student (Clem Carle), and as a follow-up to **Chapter 5** but this work is not included in this PhD thesis. Each insight relates to one research chapter and possibly one intermezzo. I will discuss my insights in the order they appear in this thesis.

In **Chapter 1** I argue that an architectural perspective provides a cohesive framework for understanding evolvability across levels of biological organisation. Organismal architecture: the structural relationships between the elements of an organism, is the feature that ultimately allows evolution to act not only as a “blind watchmaker” but also as a "reminiscing architect". Selection blindly shapes organisms to overcome selective challenges, but these adaptations become ingrained in their architecture facilitating the persistence of adaptations. This way evolution can store and re-deploy adaptations to build upon what has come before. My architectural perspective on evolvability stems from a broader idea that a mechanistic approach (i.e. paying attention to explicitly consider the mechanisms underlying biological phenomena) is key to highlighting important evolutionary dynamics. A mechanistic perspective can help reconcile the plurality of concepts around evolvability [9]. In **Chapter 2**, I argue that it is possible to categorise different determinants for evolvability on a mechanistic basis. I identify three categories of ways in which biological mechanisms can influence evolvability: by providing variation (e.g. mutation rate), by biassing the effects of variation on fitness (e.g. developmental bias), and finally by shaping the selection process (e.g. of the effect of generation time on the speed of adaptive evolution). By thinking in terms of mechanisms and how they act it becomes evident that the same mechanism can often influence evolvability in multiple ways - for example, an organism’s plastic response to the environment can affect both variation and the biassing of variation effects. **Chapter 2** also shows how the timescale as well as the type of scientific questions/approaches themselves can heavily influence the conclusions one could draw from the same data. These different conclusions, although coming from different presuppositions, are not mutually exclusive and can concur with a general explanation. I argue, therefore, that rather than attempting to subsume all observations under a single conceptual framework, it is important to clearly state the framework being used to avoid misunderstanding.

In summary, my research suggests a mechanistic and architectural framework provides a productive perspective for studying evolvability. Within this view, evolvability in all its manifestations arises from the structural relationships that connect the components of living systems across levels of organisation. An appreciation of architecture is key to developing a cohesive understanding of how evolvability contributes to the adaptive success of life on Earth. This concept helps to better appreciate the results of the simulation studies I conducted.

In **Chapter 3** I indeed observe that when interconnected together, even simple ensembles of inheritable regulatory units (genes), can easily display potential for the evolution of evolvability through natural selection. This finding is in line with pre-existing literature on Gene Regulatory Network studies that show how by accounting for complex networks of interactions between genes, evolvability can be easily observed evolving [10, 11]. However, in this case, my research brings the additional insight that architectures need not be complex to achieve this. As we show, even simple GRN systems with less than five genes and simple linear interactions can structure themselves to increase their ability to adapt to varying selective challenges. In this study, individuals achieved this result by evolving an architecture that biases the effects of random mutations toward adaptive outcomes. Small mutations can cause substantial changes in the phenotype, allowing it to “switch” from one adaptive state to another rapidly. This expedites adaptation after an environmental shift, thus increasing evolvability. Contrary to expectations, non-linear gene interactions are not required to evolve such a switch and an increase in evolvability. Thus, even architectures entailing only linear interactions can integrate information about the selective challenge into their architecture. This finding highlights how for evolvability, the existence of structural relationships between the elements of an organism may matter more than their nature (linear vs. non-linear). In **Chapter 4** I show that adaptations can evolve and endure even when selection occurs infrequently. This has two important implications. Once adaptations are obtained, they can stay ingrained in organisms for an extended period without being repeatedly selected, serving as evolutionary "memory" that can be recalled again if need be. Secondly, organisms can evolve this type of adaptation, and consequently acquire evolutionary "memory," even by just being exposed to selection rarely. This finding can help explain how organisms often display complex adaptations to environments to which apparently they have never or only very sporadically exposed (as discussed as well in Intermezzo 2). These findings in my model hold for a wide range of GRN architectures, selection event rarity, and selective challenges. They are also partially in line with previous theoretical models, which show that adaptations can be maintained for long periods in infinite populations with simple binary phenotypes. However, by including finite populations, complex multidimensional phenotypes, and more complex genomes (e.g., GRNs instead of simple genotype-to-phenotype mappings), my model yields different and new results from the previous ones as well. Despite being capable of being maintained for long periods of time in rare selection, in my model, adaptations decay very rapidly when selection is completely absent. This does not happen in previous models, where adaptations can resist for extremely long periods of time (tens of thousands of generations) under pure mutational drift. The fact that in my model adaptations decay rapidly when only mutations take place and instead persist when selection happens infrequently, shows that individuals evolve a form of robustness to mutations in response to rare selective events. The fact that in previous models individuals can simply maintain adaptation for long periods without any selective incentive, suggests that in their case robustness was innate and not evolved. Therefore once again, considering architecture (GRN vs. simple genotype-to-phenotype mappings) leads to interesting and different results than more standard modelling would predict. The evolution of robustness in response to rare selection can be interpreted as the ability of GRNs to integrate past selective events in their architecture (in this case understood as the strength of their gene-to-gene interaction) through evolution, similar to the findings in **Chapter 3**. Even very simple GRN architectures (less than 10 genes) display this ability. However, I also found that GRNs of varying sizes have different evolutionary properties. When selected against more complicated selective challenges, larger architectures show a higher level of adaptation. This happens despite these larger architectures possessing more mutable loci, and therefore being more prone to mutations that might disrupt their functioning. Our model suggests that bigger and potentially more fragile architectures are not necessarily an impairment or a cost. Evolution seems able to compensate for their potential fragility by more easily tuning them to complex tasks and making them robust thanks to the many degrees of freedom they offer with their high number of genes. Overall, these findings (chapters 3 and 4 and Intermezzo 2) highlight the importance of the structural relationships between elements of an organism in improving its ability to adapt.

In the cross-over research in Evolutionary Computation that I conducted in **Chapter 5**, I show how ideas from evolutionary biology can support research in other evolutionary fields, particularly that of Grammar Guided Genetic Programming (GGGP). In this project, I show how biasing mutation rates and effects can improve efficiency when evolution searches for adaptations. These ideas are partially inspired by the theory of Facilitated Variation from Kirschner and Gerhart [12], whose work on evolvability is influential [13] in evolutionary biology but not very well known in the field of evolutionary computation. My modelling choices in this chapter adhere to a more phenomenological approach. I do not let biased mutation effects emerge spontaneously, rather, I explicitly implement different mutation rates for tiers of genes with different effects. However, this phenomenological implementation is only possible thanks to the focus on the structural relationships of the genotype-to-phenotype map used in GGGP. This representation is embodied by a grammar that, by definition, describes the relationships between the components that participate in phenotype formation. By taking advantage of the architectural features of this grammar it is possible to efficiently assign appropriate mutation rates that allow my method to beat other state-of-the-art solutions. This shows yet again that organismal architecture is a key aspect in understanding and predicting evolutionary dynamics.

GGGP has numerous intriguing properties that make it a promising candidate for future evolutionary studies as well. The amount of potential grammar that can be designed is virtually limitless, and more importantly, the elements of the grammar itself, as well as its associated mutation rates can be readily evolved too. This opens up a wide range of applications, both practical and theoretical.

In particular, the study of evolvability can benefit from this modelling framework. Individuals can be evolved against many different well-characterised tasks (i.e. selective challenges) used in Artificial Intelligence and their phenotype can be easily interpreted thanks to the grammar determining its genotype-phenotype map. Moreover, I find the utilisation of image recognition as a selective task particularly interesting, as it can be calibrated to various degrees of complexity. This could allow researchers to study evolvability on a fine-grained scale. In a project not shown in this thesis, I (and my collaborator Pedro Carvalho) extend the study presented in **Chapter 5** to determine if our method improves adaptation not only for one but for multiple image recognition tasks. The idea is that by starting from simple image-recognition tasks and progressively increasing their complexity the biased mutation rates will allow the preservation of functional modules. Thanks to the grammar, these modules are easily identifiable in the genome, and it is possible to observe if they are redeployed for more complex challenges. Preliminary results suggest that solutions evolved with our method adapt more rapidly to new challenges (have higher evolvability) and display more streamlined and coherent phenotypes. This solution maintains high performance not only for the present selective challenge to which they are exposed but also for past ones.

In summary, my research highlights how an architectural perspective could be crucial for developing a comprehensive understanding of evolvability. The structural relationships between components at all levels of biological organisation fundamentally shape how evolvability arises and contributes to adaptive evolution. Whether considering insights on defining and reconciling evolvability, or insights derived from computational models of regulatory networks, architecture lies at the heart of evolvability. Simple network architectures can evolve complex mechanisms for biasing mutations and adapting to infrequent selection pressures. And ideas inspired by biological systems architectures can even benefit other evolutionary methods like evolutionary computation. Organismal architecture is the key to comprehending evolvability in all its diverse manifestations, from genetic evolution to non-genetic inheritance and beyond. By adopting an architectural vantage point, we gain a cohesive framework for exploring evolvability as a fundamental property of life.

6.2.1. FUTURE DIRECTIONS: STUDYING EVOLVABILITY AND NOVELTY, OPEN-ENDED EVOLUTION

The GRN models I discuss in this thesis have an important limitation when it comes to the study of evolvability: they can provide only limited insights into evolvability intended as the ability to produce evolutionary novelty. GRN models are an appropriate tool for studying "fine-tuning evolvability," or how variation can be shaped by adjusting architecture. However, these models tell us little about what might have underlined high-level evolutionary phenomena, such as the major evolutionary transitions (such as aerobic metabolism, sexual reproduction, multicellularity, bilateral symmetry, etc.). Despite allowing for unpredictable and counterintuitive outcomes, these models have a pre-established, unchangeable set of behaviours, thus making it quite impossible for them to come up with game-changing solutions. It is extremely difficult to design a simulation where GRNs that, as in **Chapter 3**, are selected to adapt to different environments, end up discovering multicellularity. Behaviours in these systems need to be pre-programmed and cannot be altered, so evolution can only fine-tune and complexify the dynamics, not revolutionise them. If one wants to observe the emergence of such phenomena the simulation needs to be appropriately “seeded” for those. I think that understanding how evolution can produce novel traits is one of the most fascinating enigmas of evolutionary studies.Understanding major transitions in evolution is a challenge for all evolutionary scientists and not only biologists. To this day, we have not yet been able to produce a model where evolution does not eventually stagnate and instead demonstrates the “open-endedness” we observe in life’s historical record.

This is one of the main interests of the research field of Artificial Life, in particular of that sub-group of scientists who are striving to obtain or detect evidence of “Open-ended evolution” [14–16]. A debate is ongoing within this multi-disciplinary community as to what is “Open-ended evolution”, how one can measure it and most importantly how one can obtain it. This debate is unknown to most biologists, as the scientists involved often hail from computational backgrounds. However, interesting insights have been garnered, and rigorous metrics of “open-endedness” are under constant development and review [17]. I think evolutionary biology should step in more decisively and bring its contribution to this discussion.

Harnessing insights from other disciplines and potentially aiding them in their endeavours could create a fertile ecosystem. Different evolutionary disciplines could help each other in speeding up progress along multiple lines of research. For instance, I think that many of the models adopted by artificial life scientists could be of great interest to biologists [18, 19]. The field of Artificial Life is a very creative modelling field, where few modelling paradigms are established, and uniqueness in mechanisms and dynamics is valued. This attitude leads to a wide variety of models, which are substantially different. Thus, when general insights are drawn from this varied bouquet of representations of evolution it makes a strong argument for their soundness. At the same time, providing more accurate insights into how biological organisms work could offer better guidance for artificial life modelling choices.

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Capturing the facets of evolvability in a mechanistic framework

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'Evolvability' – the ability to undergo adaptive evolution – is a key concept for understanding and predicting the response of biological systems to environmental change. Evolvability has various facets and is applied in many ways, easily leading to misunderstandings among researchers. To clarify matters, we first categorize the mechanisms and organismal features underlying evolvability into determinants providing variation, determinants shaping the effect of variation on fitness, and determinants shaping the selection process. Second, we stress the importance of timescale when studying evolvability. Third, we distinguish between evolvability determinants with a broad and a narrow scope. Finally, we highlight two contrasting perspectives on evolvability: general evolvability and specific evolvability. We hope that this framework facilitates communication and guides future research.

Highlights

Evolvability, the capability to undergo adaptive evolution, is determined by a staggering diversity of mechanisms and organismal features. When discussing evolvability, it is useful to distinguish three categories of determinants: those providing variation, those shaping the effect of variation on fitness, and those shaping the selection process.

Some determinants of evolvability have a broad scope in that they affect adaptive evolution across many different environments; others have a narrower scope in that they impact evolvability only with respect to particular challenges. Being explicit about the scope of evolvability determinants would largely facilitate communication across disciplines.

On different timescales, the comparison of organisms regarding their evolvability and the comparison of mechanisms regarding their effects on evolvability can lead to very different conclusions.

Evolvability is an important yet elusive concept

Understanding adaptation to changing environments is more important than ever. Climate change, antibiotic resistance, and viral vaccine evasion represent major societal challenges. Is an endangered species able to adapt to environmental change? Will a bacterial pathogen evolve antibiotic resistance? Can a virus evade vaccine-based immunization? At the core of these issues lies a common element: the capability of organisms to adapt – evolvability [1]. Evolvability research sheds new light on genomic architecture [2], the structure of regulatory networks [3,4], and many other features of **biological systems** (see Glossary). It has yielded surprising new insights, such as: adaptive evolution can proceed at a pace similar to ecological change, resulting in intricate and unexpected ecoevolutionary dynamics [5,6]; evolvability and **robustness** do not conflict but mutually reinforce each other [3,7,8]; and organisms with high evolvability can 'generalize' over environments [9,10]. Furthermore, evolvability research may add new perspectives to the formulation of a predictive theory of evolution (Box 1 and see Outstanding questions).

Evolvability is studied by diverse approaches; for instance: Johansson *et al.* [11] inspect the genetic variance–covariance matrix; Woods *et al.* [12] compare the speed of adaptation of bacterial strains; and Martín-Serra *et al.* [13] focus on morphological integration and **modularity**. These approaches, although all valid, are widely disparate: all aim to understand evolvability, yet each focuses on a different facet. This plurality is also reflected in the fact that evolvability has been defined in many different ways (Box 1). We aim to highlight the different facets of evolvability and how they relate to each other, to facilitate a more nuanced and cohesive discourse on the topic. Throughout, we define evolvability as the capability of a biological system to undergo adaptive evolution (see Box 1 for a justification).

Toward a mechanistic approach to evolvability

Evolvability is often viewed in terms of outcomes (e.g., speed of adaptation). As the same outcome can be achieved in many ways, it is useful to study evolvability by a mechanistic approach

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Box 1. Definitions of evolvability

Here we briefly discuss some definitions of evolvability, as they provide a good overview of the diversity of approaches in the field of evolvability research [63,72,73]. One important early definition revolves around the additive genetic coefficient of variation. It defines evolvability as the ability to respond to selection as governed by the presence or absence of standing genetic variation (often assessed in the G matrix) [38,58,74]. Another important perspective was given by Wagner and Altenberg [59], who made a distinction between variation and variability (i.e., the propensity of characters to vary). Evolvability is then considered not as the currently present variation but instead as the ability to generate new variation. A third, different perspective on evolvability considers the ability to generate major innovations [75,76]. For a comprehensive review of these developments, the reader is referred to [1]. Note that these definitions of evolvability are reflected in our first category of determinants, as they all view evolvability as determined by the presence or provisioning of variation.

One additional aspect of the definitions of evolvability (also called 'evolutionary potential' or 'adaptive capacity') is the relationship between variation and adaptation. While earlier treatments (as discussed in [1]) define evolvability as the ability of a biological system to evolve, irrespective of whether evolution is adaptive or not, recent definitions tend to restrict the concept to adaptive processes. For example, Payne and Wagner [14] combine the aspects of variation and adaptation when they define evolvability as '...the ability of a biological system to produce phenotypic variation that is both heritable and adaptive'. Other definitions in this vein have been provided in [7,38,60,77–81].

Following the general trend in the field, we here focus on adaptive evolution as well. When defining evolvability as the 'capability of a biological system to undergo adaptive evolution' we do not mean the presence or absence of this capability, but rather we consider its degree in a continuous fashion. Adopting an adaptive perspective does by no means imply that nonadaptive processes (e.g., genetic drift) are irrelevant; many of the determinants of evolvability reflect such processes (e.g., mutation). However, there are at least two reasons for focussing on adaptive evolution. First, many applications of evolvability (e.g., evolution of antibiotic resistance, adaptation to anthropogenic change) consider the adaptation to environmental challenges. Second, relating the rate and outcome of evolution to underlying selection pressures provides a yardstick, making it possible to differentiate between organismal and environmental features [38] and allowing comparisons across organisms. Both features are important first steps toward a predictive theory of evolution [82].

[14]: viewing evolvability not as a phenomenon *per se* but as a product of the mechanisms and organismal features that underlie it. A mechanistic perspective also clarifies discussions on the evolution of evolvability [1,14]: while questions regarding the evolution of 'the capability to undergo adaptive evolution' easily turn abstract, they become more obvious and transparent when translated into questions regarding the evolution of concrete mechanisms (e.g., the mutation rate).

Categorizing the determinants of evolvability

We refer to the mechanisms and organismal features that govern evolvability as determinants of evolvability. These affect different aspects of adaptive evolution and consequently shape evolvability in different ways. We here identify three ways in which determinants can shape evolvability, based on what aspect of adaptive evolution they affect, and categorize them accordingly (Figure 1). First, determinants may affect evolvability by providing variation. The mutation rate is the most obvious example of such a determinant [2,15–17]. Second, determinants may affect evolvability by influencing the effect of variation on fitness. For example, **developmental biases** may predispose mutations toward being beneficial [18–21]. Third, determinants may affect evolvability by shaping the selection process; shorter generation times, for example, may accelerate adaptation [22,23].

Category 1: providing variation

Heritable variation serves as the raw material for evolution. Hence, our first category refers to those mechanisms that generate and maintain variation. For example, mutations generate variation in many ways, ranging from point mutations to genome rearrangements. Interestingly, mutation rates vary widely between organisms as well as within genomes [24–26] and they can be regulated based on the environment (e.g., stress-induced mutagenesis, [27]), indicating that evolvability can evolve through the evolution of the mutation rate. Examples of determinants maintaining variation include **evolutionary capacitors** such as heat shock proteins. HSP-90 in *Saccharomyces*

Glossary

Biological system: we here define a biological system to be any biological entity that can be subject to evolution by natural selection.
Cryptic genetic variation: standing genetic variation that has little effect on phenotypic variation under normal conditions but generates variation under changed conditions. The release of this variation can facilitate (or hamper) adaptation and thus impact evolvability.
Developmental bias: the developmental mechanisms underlying a trait can introduce biases in the variation in the phenotype, even if the underlying mutations are unbiased. These biases can be (but need not be) aligned with the direction of selection, in which case they facilitate adaptive evolution.
Developmental canalization: robustness to genetic or environmental perturbations frequently exhibited by developmental systems, leading to a stable phenotypic outcome.
Evolutionary capacitor: mechanism that prevents the expression of genetic variation under some conditions, thus allowing the accumulation of cryptic genetic variation, and 'releases' this variation under other conditions, thus exposing it to selection.
Gene regulatory network (GRN) model: model that explicitly represents the genotype-to-phenotype map as a complex network of regulatory interactions between genes. GRN models have been studied extensively in the context of evolutionary developmental biology and evolvability.
Modularity: the ability of subsets of a system ('modules') to function independently of other parts of the system (see [51]). Modularity can impact evolvability in various ways – for example, independent modules can be easily combined in different ways and furthermore do not interfere with each other's functioning.
Phenotypic plasticity: the expression of different phenotypes by the same genotype in response to environmental conditions. The impact of phenotypic plasticity on evolvability is subject to much debate (e.g., [10,47–50]); their relationship is complex and not yet well understood.
Robustness: the capability of the state of a biological system to persist under (environmental or genetic) perturbation. For example, robustness may refer to

cerevisiae, for instance, acts as a chaperone protein aiding correct protein folding. Chaperoning can shield sequence mutations from selection, thus maintaining variation. This can later be released under stressful conditions [28,29]. **Developmental canalization** can affect evolvability in a similar manner, in that it allows the accumulation of **cryptic genetic variation** [30]. Furthermore, capacitors may also be behavioural in nature, as parental care and thermoregulatory behaviour also allow cryptic genetic variation to accumulate [31,32]. Horizontal gene transfer may also be viewed as a category 1 determinant, as it allows variants to be maintained that would otherwise be lost from the population; for instance, by establishing an 'accessory genome' [33,34] or through the so-called rescuable gene hypothesis [35]. Not all heritable variation is genetic: epigenetic inheritance, inheritance of environmental features, and cultural inheritance can also affect adaptive evolution [36,37]. Hence, category 1 also includes mechanisms providing non-genetic heritable variation.

the ability to maintain a certain phenotype in the face of environmental fluctuations or genetic mutations. Intuitively, one might consider evolvability (the ability to change) and robustness (the ability to withstand change) to be opposed; however, it has been shown that they can be two sides of the same coin [3,7].

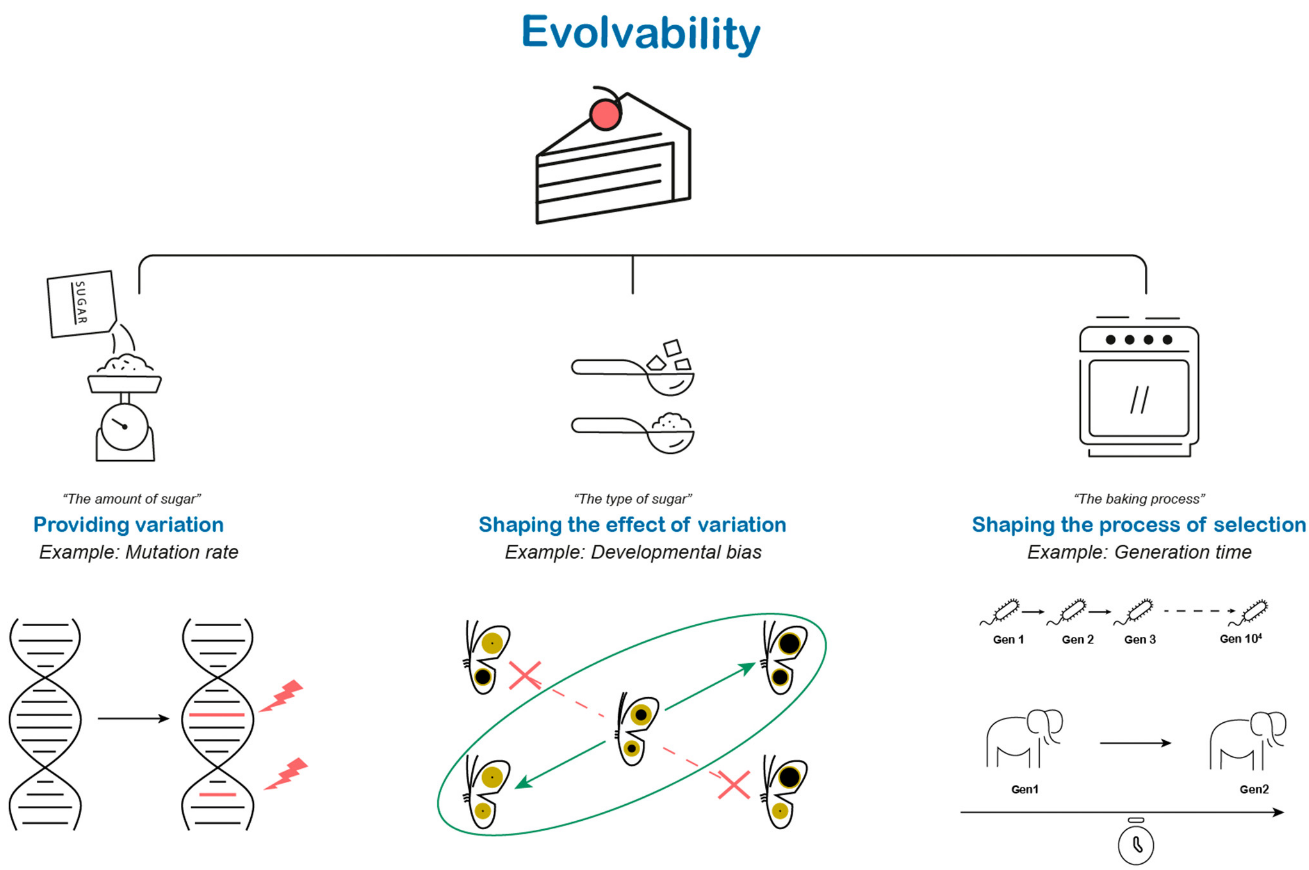

Figure 1. The way that mechanisms and organismal features affect evolvability can be classified into three categories. Each one contributes to evolvability in a different way. This can be compared with the process of baking a cake: the end result depends on several fundamentally different aspects – the amount of ingredients, the quality of ingredients, and the baking process. We suggest that evolvability is analogously affected by three different classes of determinants: those providing variation ('the amount of sugar'), those shaping the effect of variation on fitness ('the type of sugar'), and those shaping the selection process ('the baking process'). An example of a determinant providing variation is the mutation rate, where mutation encapsulates a wide variety of phenomena ranging from point mutations to genome rearrangements. Developmental biases are examples of determinants that shape the effect of variation on fitness. Consider, for instance, the developmental system underlying the eyespot pattern on the wings of the butterfly *Bicyclus anynana*. This system is organized in such a manner that mutations can easily change the colour composition of the two wing eyespots in the same direction, while mutations changing the colour composition in opposite directions are extremely rare (depiction based on [40]). Depending on whether the selective pressure favours eyespots with the same colour composition or not, this bias may facilitate or impede evolvability. Finally, an example of a determinant that shapes the selection process is generation time: a shorter generation time allows faster adaptation – in absolute time, bacteria evolve faster than elephants.

Category 2: shaping the effect of variation on fitness

The mapping from mutation to fitness is affected by a variety of mechanisms: mutations may be random with respect to the genotype, but their effects on the phenotype and consequently fitness are often not [8]. Through features such as genomic, developmental, and regulatory architecture, the genotype-to-phenotype-to-fitness map can bias the fitness effects of mutations [19,38]. Category 2 thus contains determinants that influence the effect of variation on fitness. Examples can be found in the evo-devo literature [19,21,39,40], which describes various biases introduced through the developmental process (developmental bias; for an example see the butterfly *Bicyclus anynana* in Figure 1). The effect of variation on fitness can also be biased by the genomic and regulatory architecture [4,41]. In yeast, for example, genes for which upregulation is selectively favoured in a higher-temperature environment are grouped on the same chromosome. Therefore, a duplication of this chromosome suffices to achieve upregulation of all relevant genes; without such genome organisation, many independent mutations would be required to obtain an equivalent high-temperature adaptation [42].

Category 3: shaping the selection process

Starting from the same variation, evolution can still proceed at a very different pace and/or can lead to very different outcomes. Thus, category 3 contains determinants that impact evolvability not by shaping variation, but rather by shaping how the selective process acts on this variation. Examples are organismal features influencing population structure (e.g., dispersal tendency, mating patterns), as population structure may strongly affect adaptive evolution [43]. For instance, limited dispersal is hypothesized to have aided the rapid evolution of eusociality in diverse clades of insects [44]. Two other examples of category 3 determinants are generation time and the mode of reproduction. In coevolutionary host–pathogen arms races, the shorter generation time of pathogens provides them with an evolvability advantage, as they can evolve faster per time unit than their host [23]. Considering the Red Queen hypothesis, it becomes evident that hosts need other adaptations (e.g., sexual reproduction, a variation-generating immune system) to cope with pathogens on a longer-term perspective [45]. In the coevolution of hosts and their symbionts, the rapid evolution and/or diversification of the symbiont is often not in the interest of its host. Accordingly, the hosts of various symbiotic systems reduce the symbiont's evolvability by actively interfering with the symbiont's sexual reproduction [46].

How a mechanistic categorisation aids our understanding of evolvability

Some determinants of evolvability can be classified into more than one category. This is a deliberate feature of the proposed categorisation, as it highlights that a determinant can affect evolvability in different ways. The categorisation prompts the researcher to critically consider how mechanisms and processes shape adaptive evolution. An illustrative example can be found in the literature on evolvability and **plasticity**. Some have argued that plasticity impedes evolvability: plastic responses shield organisms from selection, preventing genetic adaptation (category 3) [47]. Others have argued that plasticity allows the accumulation of cryptic genetic variation (category 1) [10,48], thus potentially enhancing evolvability, because plastic traits are expressed only under particular environmental conditions. Finally, arguments derived from **gene regulatory network (GRN) models** conclude that the evolution of plasticity can restructure the genotype–phenotype map in such a way that random mutations are more likely to produce adaptive phenotypes (category 2) [49,50]. Our categorisation of determinants thus showcases these often subtle but nevertheless crucial distinctions.

The effect of modularity on evolvability provides another example. Inspection of the underlying mechanisms reveals that modularity has not one but two impacts on evolvability. First, it facilitates innovation by allowing pre-existing modules to be combined in different configurations, thus providing variation (category 1) [51]. Second, it also allows individual modules to vary independently without affecting the functionality of the entire system (reducing antagonistic pleiotropy); this makes deleterious mutations less impactful, thus creating an adaptive bias that shapes the fitness effects of variation (category 2) [52]. This reduced impact of deleterious mutations (category 2) may also allow organisms to tolerate higher mutation rates (category 1), showing that determinants in different categories can interact in a reciprocal manner: the processes that provide variation (category 1), bias the fitness effects of variation (category 2), and shape the selection process (category 3) are not independent of each other.

Our view on the determinants of evolvability is suited to different approaches to evolution and evolvability. The first category (providing variation) contains not only mechanisms that provide new mutations, but also mechanisms that facilitate major innovations (however, the relationship between evolvability and major innovations is not yet well understood). Similarly, the second category not only considers instances of genotypic and developmental biases, but also includes broader ideas such as phenotypic accommodation and the theory of facilitated variation [20,53–55]. Finally, the third category considers not only genetic mechanisms (e.g., horizontal gene transfer, which also allows beneficial variants to spread more quickly [56]) but also – among others – niche construction, where organisms shape their own selective environment [57].

Explicitly considering timescale resolves apparent incongruencies

Determinants differ in the timescale on which they act; thus, when comparing evolvability across biological systems, the outcome is crucially dependent on timescale (Figure 2). Considering the timescale can help to resolve several apparent discrepancies.

This is exemplified by comparing determinants that provide variation (category 1): consider the impact of standing genetic variation [58] and the impact of mechanisms generating variation [59] on evolvability ([1]; Box 1). In the short term, adaptation is more strongly influenced by standing genetic variation, whereas mechanisms generating and maintaining variation are of greater significance when considering longer-term evolutionary trajectories [60].

Some approach evolvability in terms of speed of evolution [61], while others approach it in terms of the attained level of adaptation [38,62]. Both aspects are relevant [63], but they are often two

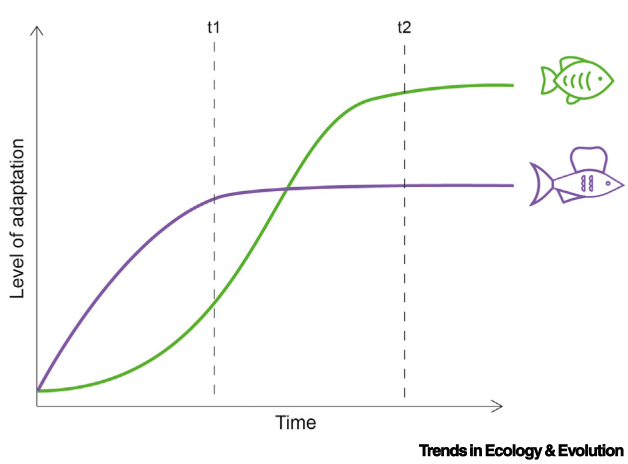

Figure 2. When studying evolvability, it is important to explicitly consider the timescale. Observing at different times can lead to different conclusions. Suppose that we observe the ability of two fish species to adapt to a new food source. Our conclusions on which of the two is more evolvable (is better able to adapt to the new selective challenge) will depend on the time at which we observe their level of adaptation. At time *t*1, the purple fish species is more adapted (and hence seems more evolvable), but at time *t*2 the green fish species is more adapted (and hence seems more evolvable). This occurs because rates of adaptation are not constant across time; thus, being explicit about the timescale of observation can resolve apparent discrepancies when comparing the evolvability of different organisms.

sides of the same coin, which becomes apparent when explicitly considering timescale. Consider, for example, different modes of inheritance. Epigenetically inherited traits can provide fast adaptation, yet this adaptation is often relatively inaccurate, given that the relative instability of epigenetic marks impedes the reliable maintenance of a certain optimal phenotype. By contrast, genetic adaptation proceeds more slowly, but in view of the high fidelity of genetic inheritance it may, in the long term, result in a higher level of adaptation. Therefore, epigenetic inheritance confers higher evolvability in the short term and genetic inheritance confers higher evolvability in the long term [36].

Another discrepancy that can be resolved by considering the timescale is the debate over the evolvability benefits of sexual reproduction, with both sexual and asexual reproduction being linked to increased evolvability [1,64]. All other things being equal, the response to selection (and hence the rate of adaptive evolution) is higher under asexual reproduction, as in the case of sexual reproduction, selection can act only on the additive component of genetic variation [65] (category 3). Therefore, asexual reproduction facilitates evolvability in the short term. In the longer term, sexual reproduction confers a higher evolvability, as the slower speed of evolution is outweighed by the ability to better explore the fitness landscape and reach global rather than local peaks. The claims that sexual reproduction increases evolvability and the claims that asexual reproduction increases evolvability can thus both be true, just at different timescales (Figure 2). Overall, the earlier examples show that the effects and relative importance of determinants vary over time. Therefore, explicit consideration of the timescale is crucial when studying evolvability.

Accounting for environmental context shows that determinants differ in scope

Evolvability is the capability to undergo adaptive evolution; it is therefore necessary to consider in relation to which environmental challenge such adaptation arises. This reveals an additional property of determinants: their scope. Some determinants affect evolvability across many different environmental challenges; we consider these to have a broad scope. For example, mutation rates impact evolvability in virtually all environments. Other determinants have a narrow scope as they shape evolvability in only a restricted set of environments. For example, the grouping of temperature-relevant genes on one chromosome in yeast [42] enhances evolvability only to a change in temperature; it does not impact adaptation to other environmental challenges.

The scope of determinants pushes the researcher to consider the range of environments in which a determinant is relevant. Determinants relevant for adaptation to one environment may not be as relevant when considering adaptation to another. For example, in the radiation of Darwin's finches, developmental biases in beak development have been implicated in their adaptation to different seed sizes [66,67]. However, evolvability regarding beak shape will not be relevant with regard to other environmental challenges, such as temperature regulation or predator escape. By contrast, a higher mutation rate will affect evolutionary adaptation with regard to many different environmental challenges.

Two perspectives on evolvability

A very different distinction does not refer to the determinants of evolvability, but to the scholars studying evolvability. Depending on their scientific discipline, research question, or model system, scholars differ in whether they view evolvability as 'general' or 'specific' with regard to environmental challenges (Figure 3). Scientists adopting a specific evolvability perspective refer to the capability of a biological system to undergo adaptive evolution to a specific environment or a specific challenge; the main focus therefore is on how well a biological system can meet a specific selective target. This perspective is useful when studying adaptation to a particular challenge; for example, when exploring the ability of bacteria to evolve resistance to a particular antibiotic,

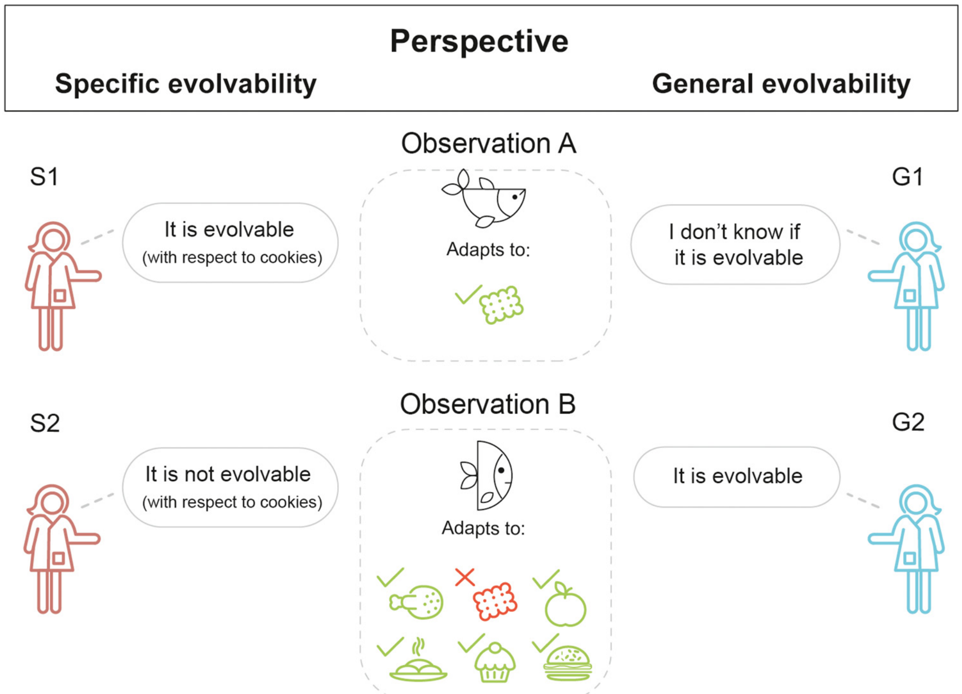

Figure 3. Specific and general evolvability represent two different perspectives on evolvability. This influences questions and the interpretation of results in evolvability research. Different scientists can reach different conclusions from the same observations. This figure illustrates how scientists with different perspectives (on the left: specific evolvability; on the right: general evolvability) interpret the same observations regarding adaptation very differently. Observation A: a fish species can easily adapt to using cookies as a food source. From the specific evolvability perspective (S1), this is interpreted as high evolvability with respect to cookies. From the general evolvability perspective (G1), it is not possible to draw conclusions since no information is available regarding the ability to adapt to other food sources (environments). Observation B: a fish species cannot adapt to using cookies as food source, but (e.g., due to modular mouth parts) can undergo adaptation to a wide range of other food sources (environments). From a specific evolvability perspective (S2), this is interpreted as a lack of evolvability with regard to cookies. From a general evolvability perspective (G2), the ability to adapt to a wide range of different environments (ability to deal with dietary shifts) indicates high evolvability. Notice that the two perspectives characterise scholars of evolvability rather than the determinants of evolvability. The two perspectives should therefore not be confused with determinants acting at short versus long timescales or with determinants having narrow versus broad scope.

the ability of a virus to evolve resistance to a vaccine, or the ability of an endangered species to evolve adaptations to a specific anthropogenic threat [68,69]. By contrast, scientists adopting a general evolvability perspective view evolvability as the capability of a biological system to adapt to a wide spectrum of environments or of challenges, thus effectively considering evolvability irrespective of the environmental context. This perspective is useful when considering adaptation to unpredictable environments and is frequently used in studies exploring the link between evolvability and diversification [62,70,71].

While specific and general evolvability are both useful conceptualizations of evolvability, insights gained from one do not necessarily translate into insights about the other. Depending on the chosen perspective, the same observation can lead to different conclusions (Figure 3); it informs what questions are asked and affects how results are interpreted. Consider a population that is able to adapt rapidly to a specific challenge, such as a bacterial strain quickly evolving resistance to a particular antibiotic. From the perspective of specific evolvability, this strain has a high evolvability, while viewed from the perspective of general evolvability this single instance of rapid adaptation says nothing about the ability of the strain to adapt to other challenges (heat stress, pH stress, etc.). The distinction between general and specific evolvability should not be confused with the scope of a determinant: the latter is a property of a determinant, whereas the former concerns two different ways of viewing evolvability.

Concluding remarks

Evolvability is an intricate concept with many facets. Different facets are at centre stage in different lines of research. Furthermore, evolvability is conceptualized in two different ways: specific and general evolvability. Being aware of these differences is crucial to foster an integrated and structured view on evolvability research.

Throughout we argue that evolvability should not be studied as a phenomenon *per se* but as a product of the mechanisms underlying it. Moreover, it is useful to clearly distinguish between determinants that provide variation, shape the effect of variation on fitness, and shape the selection process. A structured mechanistic approach clarifies debates in the literature and provides a sound basis for studying the evolution of evolvability.

Evolvability cannot be quantified by a single number. Both speed of evolution and level of adaptation are relevant, but they are not independent. Scholars should explicitly consider this when conducting evolvability research.

We hope that the proposed mechanistic approach facilitates communication across disciplines, helps to address major questions regarding evolvability (see Outstanding questions), and provides guidelines for the design of future studies on evolvability.

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Declaration of interests

No interests are declared.

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Outstanding questions

What mechanisms are most important in determining evolvability?

While we are slowly gaining a better understanding of the mechanisms that underlie evolvability, the relative importance of these mechanisms is not always apparent. Which mechanisms are the key drivers of evolvability and its evolution? How does this depend on the ecoevolutionary context?

How do plasticity and evolvability interact?

The relationship between evolvability and plasticity is currently hotly debated and should remain a topic of future study. This discussion includes various questions and concepts, such as the 'plasticity first hypothesis', the 'flexible stem hypothesis', and the question of whether genes are ' leaders or followers' in evolution.

Does evolvability evolve and if so how?

A mechanistic perspective may help in understanding the evolution of evolvability, as the mechanisms underlying evolvability can clearly evolve. Still, the question remains about whether and how organismal features that are mainly relevant on a long-term perspective (like the ability to adapt to novel environmental challenges) can be shaped by the myopic process of natural selection. Can evolvability be the target of selection or is it the by-product of other processes?

Can evolvability help in the formulation of a predictive theory of evolution?

The question of whether or to what extent evolution is predictable has fascinated evolutionary biologists for decades. Evolvability research may contribute new perspectives to this question: Understanding evolvability may be a first step toward a probabilistic prediction of evolutionary trajectories.

How can practical applications benefit from evolvability research?

While it is clear that evolvability is relevant for major societal problems, such as the evolution of antibiotic

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resistance and the implications of climate change, the full benefits of an explicit consideration of evolvability – as well as the methodology for doing so – remain to be explored.

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